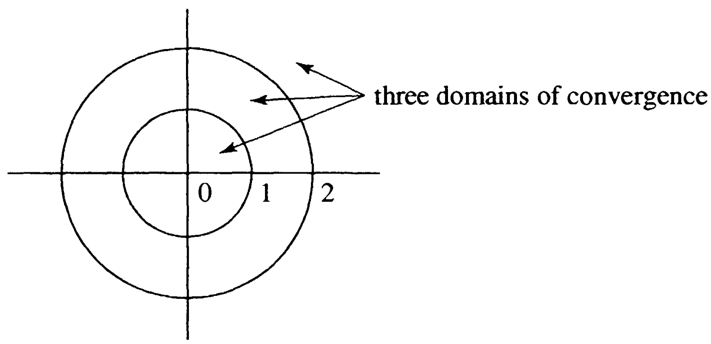
Example. The function $f(z) = 1/(z-1)(z-2)$ has three Laurent decompositions centered at 0. One represents the function in the punctured disk $\{0 < |z| < 1\}$, one in the annulus $\{1 < |z| < 2\}$, and one in the exterior domain $\{2 < |z| < \infty\}$. Since the function is already analytic in the disk $\{|z| < 1\}$, the decomposition in the punctured disk is given by $f_0(z) = f(z)$ and $f_1(z) = 0$. Since the function is analytic at ∞ and vanishes there, its Laurent decomposition with respect to the exterior domain $\{2 < |z| < \infty\}$ is given by $f(z) = f_1(z)$ and $f_0(z) = 0$. The only nontrivial Laurent decomposition is with respect to the annulus $\{1 < |z| < 2\}$. To see what it is, we consider the partial fractions decomposition

$$\frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}.$$

The summand $f_0(z) = 1/(z-2)$ is analytic for $|z| < 2$, and the summand $f_1(z) = -1/(z-1)$ is analytic for $|z| > 1$ and it vanishes at ∞. Thus this partial fractions decomposition coincides with the Laurent decomposition with respect to the annulus.



Example. The function $1/\sin z$ has a Laurent decomposition with respect to each annulus $\{n\pi < |z| < (n+1)\pi\}$, for $n = 0, 1, 2, \ldots$. We will see how to obtain the Laurent decompositions in the next section.

Suppose now $f(z) = f_0(z) + f_1(z)$ is the Laurent decomposition for a function analytic for $\rho < |z - z_0| < \sigma$. We can express $f_0(z)$ as a power series in $z - z_0$,

$$f_0(z) = \sum_{k=0}^{\infty} a_k (z-z_0)^k, \qquad |z - z_0| < \sigma,$$

where the series converges absolutely, and for any $s < \sigma$ it converges uniformly for $|z - z_0| \leq s$. Further, we can also express $f_1(z)$ as a series of negative powers of $z - z_0$, with zero constant term, since $f_1(z)$ tends to 0 at ∞,

$$f_1(z) = \sum_{k=-\infty}^{-1} a_k (z-z_0)^k, \qquad |z - z_0| > \rho.$$

This series converges absolutely, and for any $r > \rho$ it converges uniformly for $|z - z_0| \geq r$. If we add the two series, we obtain a two-tailed expansion for $f(z)$,

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k, \qquad \rho < |z - z_0| < \sigma, \tag{1.3}$$

that converges absolutely, and that converges uniformly for $r \leq |z-z_0| \leq s$. The series (1.3) is called the **Laurent series expansion** of $f(z)$ with respect to the annulus $\rho < |z - z_0| < \sigma$.

To obtain a formula for the coefficients in the expansion, we divide $f(z)$ by $(z - z_0)^{n+1}$ in (1.3) and integrate around the circle $\{|z - z_0| = r\}$. Since the series converges uniformly on the circle, we can interchange the summation and integration. The result is

$$\begin{aligned}\oint_{|z-z_0|=r} \frac{1}{(z-z_0)^{n+1}} f(z)\, dz &= \oint \frac{1}{(z-z_0)^{n+1}} \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k \, dz \\ &= \sum_{k=-\infty}^{\infty} a_k \oint_{|z-z_0|=r} (z - z_0)^{k-n-1}\, dz.\end{aligned}$$

The integral of $(z - z_0)^m$ is $2\pi i$ if $m = -1$, otherwise zero, so all the terms in the series disappear except one, and the series reduces to $2\pi i a_n$. Thus

$$a_n = \frac{1}{2\pi i} \oint_{|z-z_0|=r} \frac{f(z)}{(z-z_0)^{n+1}} dz, \qquad -\infty < n < \infty. \tag{1.4}$$

Note that this formula for a_n coincides with the usual formula for the power series coefficients in the case that $f(z)$ is analytic for $|z - z_0| < \sigma$. In the case that $f(z)$ is analytic at ∞ the formula for a_n agrees with the formula given in Section V.5.

We summarize our results in the following theorem.

Theorem (Laurent Series Expansion). *Suppose* $0 \leq \rho < \sigma \leq \infty$, *and suppose* $f(z)$ *is analytic for* $\rho < |z - z_0| < \sigma$. *Then* $f(z)$ *has a Laurent expansion* (1.3) *that converges absolutely at each point of the annulus, and that converges uniformly on each subannulus* $r \leq |z - z_0| \leq s$, *where* $\rho < r < s < \sigma$. *The coefficients are uniquely determined by* $f(z)$, *and they are given by* (1.4) *for any fixed* r, $\rho < r < \sigma$.

Example. To expand the function $f(z) = 1/(z-1)(z-2)$ in a Laurent series centered at $z = 0$ and converging in the annulus $\{1 < |z| < 2\}$, we expand each of the partial fractions in a geometric series,

$$\frac{1}{z-2} = -\frac{1}{2}\frac{1}{1-z/2} = -\frac{1}{2}\left(1+\frac{z}{2}+\frac{z^2}{4}+\cdots\right),$$
$$\frac{1}{z-1} = \frac{1}{z}\frac{1}{1-1/z} = \frac{1}{z}+\frac{1}{z^2}+\frac{1}{z^3}+\cdots.$$

This leads to the Laurent series representation

$$f(z) = \sum_{k=-\infty}^{\infty} a_k z^k, \qquad 1<|z|<2,$$

where $a_k=-1$ if $k<0$, and $a_k=-1/2^{k+1}$ if $k\ge 0$.

Example. The function $f(z)=1/(z-1)(z-2)$ can also be expanded in a Laurent series centered at $z=1$, convergent in the punctured disk $\{0<|z-1|<1\}$. Again we rely upon a geometric series,

$$\frac{1}{z-2} = -\frac{1}{1-(z-1)} = -\sum_{k=0}^{\infty}(z-1)^k, \qquad |z-1|<1,$$

to obtain

$$\begin{aligned}\frac{1}{(z-1)(z-2)} &= -\frac{1}{z-1}-1-(z-1)-(z-1)^2-(z-1)^3-\cdots \\ &= -\sum_{k=-1}^{\infty}(z-1)^k, \qquad 0<|z-1|<1.\end{aligned}$$

The tail of the series (1.3) with the positive powers of $z-z_0$ converges on the largest open disk centered at z_0 to which $f_0(z)$ extends to be analytic, while the tail of the series with the negative powers of $z-z_0$ converges on the largest exterior domain of the form $\{|z-z_0|>\tau\}$ to which $f_1(z)$ extends analytically. Thus the largest open domain on which the full Laurent series (1.3) converges is the largest open annular set centered at z_0 containing the annulus $\{\rho<|z-z_0|<\sigma\}$ to which $f(z)$ extends analytically. This annular set might extend to z_0 or to ∞, to be a punctured disk or a full disk, or a punctured complex plane or the full complex plane.

Exercise. Consider the Laurent series for $f(z)=(z^2-\pi^2)/\sin z$ that is centered at 0 and that converges for $|z|=1$. What is the largest open set on which the series converges?

Solution. Since $\sin z$ has a simple zero at π, the function $(\sin z)/(z-\pi)$ extends to be analytic and nonzero at $z=\pi$. Hence $(z^2-\pi^2)/\sin z$ extends to be analytic at $z=\pi$. Similarly, it extends to be analytic at $z=-\pi$. (We say that the singularities at $z=\pm\pi$ are "removable.") The function tends to ∞ at $z=0$ and at $z=\pm 2\pi$. Thus the largest annular domain containing the circle $\{|z|=1\}$ to which the function extends analytically is the punctured disk $\{0 < |z| < 2\pi\}$. This is then the largest open set on which the series converges.

Exercises for VI.1

1. Find all possible Laurent expansions centered at 0 of the following functions:
   (a) $\dfrac{1}{z^2 - z}$ (b) $\dfrac{z-1}{z+1}$ (c) $\dfrac{1}{(z^2-1)(z^2-4)}$

2. For each of the functions in Exercise 1, find the Laurent expansion centered at $z = -1$ that converges at $z = \frac{1}{2}$. Determine the largest open set on which each series converges.

3. Recall the power series for the Bessel function $J_n(z)$, $n \geq 0$, given in Exercise V.4.11, and define $J_{-n}(z) = (-1)^n J_n(z)$. For fixed $w \in C$, establish the Laurent series expansion

$$\exp\left[\frac{w}{2}(z - 1/z)\right] = \sum_{n=-\infty}^{\infty} J_n(w) z^n, \qquad 0 < |z| < \infty.$$

   From the coefficient formula (1.4), deduce that

$$J_n(z) = \frac{1}{2\pi}\int_0^{2\pi} e^{i(n\theta - z\sin\theta)}\, d\theta, \qquad z \in \mathbb{C}.$$

   Remark. This Laurent expansion is called the **Schlömilch formula.**

4. Suppose that $f(z) = f_0(z) + f_1(z)$ is the Laurent decomposition of an analytic function $f(z)$ on the annulus $\{A < |z| < B\}$. Show that if $f(z)$ is an even function, then $f_0(z)$ and $f_1(z)$ are even functions, and the Laurent series expansion of $f(z)$ has only even powers of z. Show that if $f(z)$ is an odd function, then $f_0(z)$ and $f_1(z)$ are odd functions, and the Laurent series expansion of $f(z)$ has only odd powers of z.

5. Suppose $f(z)$ is analytic on the punctured plane $D = \mathbb{C}\backslash\{0\}$. Show that there is a constant c such that $f(z) - c/z$ has a primitive in D. Give a formula for c in terms of an integral of $f(z)$.

6. Fix an annulus $D = \{a < |z| < b\}$, and let $f(z)$ be a continuous function on its boundary ∂D. Show that $f(z)$ can be approximated uniformly on ∂D by polynomials in z and $1/z$ if and only if $f(z)$ has a continuous extension to the closed annulus $D \cup \partial D$ that is analytic on D.

7. Show that a harmonic function u on an annulus $\{A < |z| < B\}$ has a unique expansion

$$u(re^{i\theta}) = \sum_{n=-\infty}^{\infty} a_n r^n \cos(n\theta) + \sum_{n\neq 0} b_n r^n \sin(n\theta) + c \log r,$$

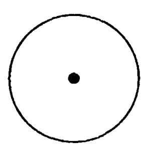
which is uniformly convergent on each circle in the annulus. Show that for each r, $A < r < B$, the coefficients a_n, b_n, and c satisfy

$$\begin{aligned} a_n r^n + a_{-n} r^{-n} &= \frac{1}{\pi}\int_{-\pi}^{\pi} u\left(re^{i\theta}\right)\cos(n\theta)d\theta, && n \neq 0, \\ b_n r^n - b_{-n} r^{-n} &= \frac{1}{\pi}\int_{-\pi}^{\pi} u\left(re^{i\theta}\right)\sin(n\theta)d\theta, && n \neq 0, \\ a_0 + c\log r &= \frac{1}{2\pi}\int_{-\pi}^{\pi} u\left(re^{i\theta}\right)d\theta. \end{aligned}$$

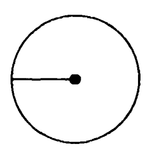
Hint. Use a decomposition of the form $u = \operatorname{Re} f + c \log|z|$, where f is analytic on the annulus.

2. Isolated Singularities of an Analytic Function

A point z_0 is an **isolated singularity** of $f(z)$ if $f(z)$ is analytic in some punctured disk $\{0 < |z - z_0| < r\}$ centered at z_0. For example, the function $1/z$ has an isolated singularity at $z = 0$, while $1/\sin z$ has isolated singularities at each of the points $z = 0, \pm\pi, \pm 2\pi, \ldots$. The functions $\sqrt{z}$ and $\log z$ do not have isolated singularities at $z = 0$; they cannot be defined even continuously on any punctured disk centered at 0.



isolated



not isolated

Suppose that $f(z)$ has an isolated singularity at z_0. Then $f(z)$ has a Laurent series expansion

$$f(z) = \sum_{k=-\infty}^{\infty} a_k (z - z_0)^k, \qquad 0 < |z - z_0| < r. \tag{2.1}$$

We classify the isolated singularity at z_0 as one of three types according to whether no negative powers of $z - z_0$ appear in the expansion, or at least one but only finitely many negative powers appear, or infinitely many negative powers appear. These are three mutually exclusive cases that cover all possibilities. We discuss each of these cases in turn, and we prove one theorem for each case.

The isolated singularity of $f(z)$ at z_0 is defined to be a **removable singularity** if $a_k = 0$ for all $k < 0$. In this case the Laurent series (2.1) becomes a power series

$$f(z) = \sum_{k=0}^{\infty} a_k(z-z_0)^k, \qquad 0 < |z-z_0| < r.$$

If we define $f(z_0) = a_0$, the function $f(z)$ becomes analytic on the entire disk $\{|z-z_0| < r\}$.

Example. The function $(\sin z)/z$ has an isolated singularity at $z = 0$, where it is not defined. From the power series expansion of $\sin z$ we obtain the Laurent series expansion

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots,$$

valid for all z. We may extend $(\sin z)/z$ to be an entire function by defining $(\sin z)/z$ to be 1 at $z = 0$.

If $f(z)$ has a removable singularity at z_0, then $f(z)$ is bounded near z_0. There is a converse statement, which provides a useful criterion for determining whether a singularity is removable.

Theorem (Riemann's Theorem on Removable Singularities). *Let z_0 be an isolated singularity of $f(z)$. If $f(z)$ is bounded near z_0, then $f(z)$ has a removable singularity at z_0.*

To see this, we expand $f(z)$ in a Laurent series (2.1) and use the formula (1.4) for the coefficients given in the preceding section. Suppose $|f(z)| \le M$ for z near z_0, and let $r > 0$ be small. Using the ML-estimate to estimate the integral in (1.4), we obtain

$$|a_n| \le \frac{1}{2\pi}\frac{M}{r^{n+1}}(2\pi r) = \frac{M}{r^n}.$$

If $n < 0$, the right-hand side tends to 0 as $r \to 0$. We conclude that $a_n = 0$ for $n < 0$, and consequently the singularity at z_0 is removable.

The isolated singularity of $f(z)$ at z_0 is defined to be a **pole** if there is $N > 0$ such that $a_{-N} \neq 0$ but $a_k = 0$ for all $k < -N$. The integer N is the **order** of the pole. In this case the Laurent series (2.1) becomes

$$f(z) = \sum_{k=-N}^{\infty} a_k(z-z_0)^k = \frac{a_{-N}}{(z-z_0)^N} + \cdots + \frac{a_{-1}}{z-z_0} + a_0 + a_1(z-z_0) + \cdots.$$

The sum of the negative powers,

$$P(z) = \sum_{k=-N}^{-1} a_k(z-z_0)^k = \frac{a_{-N}}{(z-z_0)^N} + \cdots + \frac{a_{-1}}{z-z_0},$$

is called the **principal part** of $f(z)$ at the pole z_0. The principal part $P(z)$ coincides with the summand $f_1(z)$ in the Laurent decomposition $f(z) = f_0(z)+f_1(z)$ given in the preceding section. The bad behavior of $f(z)$ at z_0 is incorporated into $P(z)$, in the sense that $f(z) - P(z)$ is analytic at z_0.

A pole of order one is called a **simple pole**, and a pole of order two is called a **double pole**. Thus $1/z$ has a simple pole at $z = 0$, and $1/(z-i)^2$ has a double pole at $z = i$.

Theorem. *Let z_0 be an isolated singularity of $f(z)$. Then z_0 is a pole of $f(z)$ of order N if and only if $f(z) = g(z)/(z-z_0)^N$, where $g(z)$ is analytic at z_0 and $g(z_0) \neq 0$.*

The proof is straightforward. Suppose that $f(z)$ has a pole of order N at z_0, and that $f(z)$ has the above Laurent series. Then the power series $a_{-N}+a_{-N+1}(z-z_0)+a_{-N+2}(z-z_0)^2+\cdots$ converges to a function $g(z)$ that is analytic at z_0 and satisfies $f(z) = g(z)/(z-z_0)^N$, and further, $g(z_0) = a_{-N} \neq 0$. Conversely, if $g(z)$ is analytic at z_0 and satisfies $g(z_0) \neq 0$, then $f(z) = g(z)/(z-z_0)^N$ has Laurent series with leading term $g(z_0)/(z-z_0)^N$, so that $f(z)$ has a pole of order N at z_0.

Theorem. *Let z_0 be an isolated singularity of $f(z)$. Then z_0 is a pole of $f(z)$ of order N if and only if $1/f(z)$ is analytic at z_0 and has a zero of order N.*

Again the proof is easy. Suppose $f(z)$ has a pole of order N at z_0. Let $g(z) = (z-z_0)^N f(z)$ be as above. Since $g(z_0) \neq 0$, the function $h(z) = 1/g(z)$ is analytic at z_0 and satisfies $h(z_0) \neq 0$. Thus $1/f(z) = (z-z_0)^N h(z)$ has a zero of order N at z_0. This argument is also reversible. If $1/f(z)$ has a zero of order N at z_0, then $1/f(z) = (z-z_0)^N h(z)$ for some analytic function $h(z)$ satisfying $h(z_0) \neq 0$, and then $g(z) = 1/h(z)$ is analytic and nonzero at z_0, so $f(z) = g(z)/(z-z_0)^N$ has a pole of order N at z_0.

Example. The function $1/\sin z$ has poles at each of the zeros of $\sin z$. Since the zeros of $\sin z$ are simple, they are simple poles for $1/\sin z$.

Exercise. Consider the Laurent series expansion for $1/\sin z$ that converges on the circle $\{|z| = 4\}$. Find the coefficients a_0, a_{-1}, a_{-2}, and a_{-3} of 1, $1/z$, $1/z^2$, and $1/z^3$, respectively. Determine the largest open set on which the series converges.

Solution. The only zeros of $\sin z$ are at the integral multiples of π. These are then the only singularities of $1/\sin z$, and they are all simple poles. The largest open annular set containing the circle and to which $1/\sin z$ extends analytically is then the annulus $\{\pi < |z| < 2\pi\}$. This annulus is then the largest open set on which the Laurent series converges. From the expansion $\sin z = z + \mathcal{O}(z^3)$ near $z = 0$, we see that

$$\frac{1}{\sin z} = \frac{1}{z} + \text{analytic}$$

near $z = 0$, and $1/\sin z - 1/z$ is analytic at $z = 0$. Similarly, from the expansion $\sin z = -(z - \pi) + \mathcal{O}((z - \pi)^3)$ at $z = \pi$, we see that

$$\frac{1}{\sin z} = -\frac{1}{z - \pi} + \text{analytic}$$

near $z = \pi$, and $1/\sin z + 1/(z - \pi)$ is analytic at $z = \pi$. By the same token, $1/\sin z + 1/(z + \pi)$ is analytic at $z = -\pi$. We conclude that if

$$f_1(z) = \frac{1}{z + \pi} + \frac{1}{z - \pi} - \frac{1}{z},$$

then $f_0(z) = 1/\sin z - f_1(z)$ is analytic for $|z| < 2\pi$. Thus $1/\sin z = f_0(z) + f_1(z)$ is the Laurent decomposition of $1/\sin z$. To obtain the negative powers of the Laurent expansion, we expand $f_1(z)$ in a series of descending powers of z, using the geometric series expansion. This is done most easily by combining the first two summands,

$$f_1(z) = \frac{2z}{z^2 - \pi^2} - \frac{1}{z} = -\frac{1}{z} + \frac{2}{z}\sum_{k=0}^{\infty} \frac{\pi^{2k}}{z^{2k}} = \frac{1}{z} + \sum_{k=1}^{\infty} \frac{2\pi^{2k}}{z^{2k+1}}.$$

Note that all the even powers of z disappear, as they must, since $1/\sin z$ is an odd function. We read off the coefficients $a_{-1} = 1$ and $a_{-3} = 2\pi^2$.

We say that a function $f(z)$ is **meromorphic** on a domain D if $f(z)$ is analytic on D except possibly at isolated singularities, each of which is a pole. Sums and products of meromorphic functions are meromorphic. Quotients of meromorphic functions are meromorphic, provided that the denominator is not identically zero. Note also that if there are infinitely many poles of $f(z)$ in D, then we can arrange them in a sequence that accumulates only at the boundary of D. Otherwise, there would be a point of accumulation in D of the poles of $f(z)$, and this point would not be an isolated singularity of $f(z)$.

Example. The function $1/\sin z$ is meromorphic on the entire complex plane. As another example, let $R(z)$ be any rational function. We can express $R(z)$ as a quotient of polynomials in the form

$$R(z) = c\frac{(z - \zeta_1)^{m_1} \cdots (z - \zeta_k)^{m_k}}{(z - z_1)^{n_1} \cdots (z - z_l)^{n_l}},$$

where the ζ_j's and z_j's are all distinct. Evidently, $R(z)$ has a zero of order m_j at each ζ_j and a pole of order n_j at each z_j. Thus $R(z)$ is meromorphic on the entire complex plane.

We add one more useful characterization of poles.

Theorem. *Let z_0 be an isolated singularity of $f(z)$. Then z_0 is a pole if and only if $|f(z)| \to \infty$ as $z \to z_0$.*

One direction of the theorem is trivial. If $f(z)$ has a pole of order N at z_0, then $g(z) = (z - z_0)^N f(z)$ is analytic and nonzero at z_0, so that

$$|f(z)| = |z - z_0|^{-N}|g(z)| \to \infty$$

as $z \to z_0$. For the converse, we use Riemann's theorem on removable singularities. Suppose $|f(z)| \to \infty$ as $z \to z_0$. Then $f(z) \neq 0$ for z near z_0, so that $h(z) = 1/f(z)$ is analytic in some punctured neighborhood of z_0. Further, $h(z) \to 0$ as $z \to z_0$. By Riemann's theorem, $h(z)$ extends to be analytic at z_0, and moreover, $h(z_0) = 0$. If N is the order of the zero of $h(z)$ at z_0, then $f(z) = 1/h(z)$ has a pole of order N at z_0.

The isolated singularity of $f(z)$ at z_0 is defined to be an **essential singularity** if $a_k \neq 0$ for infinitely many $k < 0$. Thus an isolated singularity that is neither removable nor a pole is declared to be essential.

Example. The Laurent expansion of $e^{1/z}$ at $z = 0$ is given by

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!}\frac{1}{z^2} + \frac{1}{3!}\frac{1}{z^3} + \cdots, \qquad z \neq 0.$$

Since infinitely many negative powers of z appear in the expansion, the isolated singularity at $z = 0$ is essential. That the singularity is essential can also be seen from the behavior of $e^{1/z}$ as $z \to 0$. Since $e^{1/x} \to +\infty$ as $x > 0$ tends to 0, the singularity is not removable. And since $e^{1/(iy)}$ has unit modulus, the modulus of $e^{1/z}$ does not tend to $+\infty$ as z tends to 0 along the imaginary axis, and the singularity is not a pole. The only remaining possibility is that the singularity is essential.

At an essential singularity, the values of $f(z)$ cluster towards the entire complex plane. That is the content of the following theorem.

Theorem (Casorati-Weierstrass Theorem). *Suppose z_0 is an essential isolated singularity of $f(z)$. Then for every complex number w_0, there is a sequence $z_n \to z_0$ such that $f(z_n) \to w_0$.*

We argue the contrapositive. Suppose that there is some complex number w_0 that is not a limit of values of $f(z)$ as above. Then there is some small $\varepsilon > 0$ such that $|f(z) - w_0| > \varepsilon$ for all z near z_0. Hence $h(z) = 1/(f(z) - w_0)$ is bounded near z_0. By Riemann's theorem, $h(z)$ has a removable singularity at z_0. Hence $h(z) = (z - z_0)^N g(z)$ for some $N \geq 0$ and some analytic function $g(z)$ satisfying $g(z_0) \neq 0$. Thus $f(z) - w_0 = 1/h(z) = (z - z_0)^{-N}(1/g(z))$, where $1/g(z)$ is analytic at z_0. If $N = 0$, $f(z)$ extends to be analytic at z_0, while if $N > 0$, $f(z)$ has a pole of order N at z_0. This establishes the theorem.

Later we will prove Picard's theorem, that if $f(z)$ is an analytic function with an essential isolated singularity at z_0, then for all complex numbers w_0 with possibly one exception, there is a sequence $z_n \to z_0$ such that $f(z_n) = w_0$. The function $f(z) = e^{1/z}$, which omits the value $w = 0$, shows that we must allow for the exceptional point.

Exercises for VI.2

1. Find the isolated singularities of the following functions, and determine whether they are removable, essential, or poles. Determine the order of any pole, and find the principal part at each pole.
   (a) $z/(z^2-1)^2$ (d) $\tan z = \dfrac{\sin z}{\cos z}$ (g) $\operatorname{Log}\left(1-\dfrac{1}{z}\right)$
   (b) $\dfrac{ze^z}{z^2-1}$ (e) $z^2 \sin\left(\dfrac{1}{z}\right)$ (h) $\dfrac{\operatorname{Log} z}{(z-1)^3}$
   (c) $\dfrac{e^{2z}-1}{z}$ (f) $\dfrac{\cos z}{z^2-\pi^2/4}$ (i) $e^{1/(z^2+1)}$

2. Find the radius of convergence of the power series for the following functions, expanded about the indicated point.
   (a) $\dfrac{z-1}{z^4-1}$, about $z = 3+i$, (c) $\dfrac{z}{\sin z}$, about $z = \pi i$,
   (b) $\dfrac{\cos z}{z^2-\pi^2/4}$, about $z = 0$, (d) $\dfrac{z^2}{\sin^3 z}$, about $z = \pi i$.

3. Consider the function $f(z) = \tan z$ in the annulus $\{3 < |z| < 4\}$. Let $f(z) = f_0(z) + f_1(z)$ be the Laurent decomposition of $f(z)$, so that $f_0(z)$ is analytic for $|z| < 4$, and $f_1(z)$ is analytic for $|z| > 3$ and vanishes at ∞. (a) Obtain an explicit expression for $f_1(z)$. (b) Write down the series expansion for $f_1(z)$, and determine the largest domain on which it converges. (c) Obtain the coefficients a_0, a_1, and a_2 of the power series expansion of $f_0(z)$. (d) What is the radius of convergence of the power series expansion for $f_0(z)$?

4. Suppose $f(z)$ is meromorphic on the disk $\{|z| < s\}$, with only a finite number of poles in the disk. Show that the Laurent decomposition of $f(z)$ with respect to the annulus $\{s - \varepsilon < |z| < s\}$ has the form $f(z) = f_0(z) + f_1(z)$, where $f_1(z)$ is the sum of the principal parts of $f(z)$ at its poles.

5. By estimating the coefficients of the Laurent series, prove that if z_0 is an isolated singularity of f, and if $(z-z_0)f(z) \to 0$ as $z \to z_0$, then z_0 is removable. Give a second proof based on Morera's theorem.

6. Show that if $f(z)$ is continuous on a domain D, and if $f(z)^8$ is analytic on D, then $f(z)$ is analytic on D.

7. Show that if z_0 is an isolated singularity of $f(z)$, and if $(z-z_0)^N f(z)$ is bounded near z_0, then z_0 is either removable or a pole of order at most N.

8. A meromorphic function f at z_0 is said to have **order** N at z_0 if $f(z) = (z - z_0)^N g(z)$ for some analytic function g at z_0 such that $g(z_0) \neq 0$. The order of the function 0 is defined to be $+\infty$. Show that
(a) $\mathrm{order}(fg, z_0) = \mathrm{order}(f, z_0) + \mathrm{order}(g, z_0)$,
(b) $\mathrm{order}(1/f, z_0) = -\mathrm{order}(f, z_0)$,
(c) $\mathrm{order}(f + g, z_0) \geq \min\{\mathrm{order}(f, z_0), \mathrm{order}(g, z_0)\}$.
Show that strict inequality can occur in (c), but that equality holds in (c) whenever f and g have different orders at z_0.

9. Recall that "$f(z) = \mathcal{O}(h(z))$ as $z \to z_0$" means that there is a constant C such that $|f(z)| \leq C|h(z)|$ for z near z_0. Show that if z_0 is an isolated singularity of an analytic function $f(z)$, and if $f(z) = \mathcal{O}((z-z_0)^m)$ as $z \to z_0$, then the Laurent coefficients of $f(z)$ are 0 for $k < m$, that is, the Laurent series of $f(z)$ has the form
$$f(z) \;=\; a_m(z - z_0)^m + a_{m+1}(z - z_0)^{m+1} + \cdots .$$
Remark. This shows that the use of the notation $\mathcal{O}(z^m)$ in Section V.6 is consistent.

10. Show that if $f(z)$ and $g(z)$ are analytic functions that both have the same order $N \geq 0$ at z_0, then
$$\lim_{z\to z_0} \frac{f(z)}{g(z)} \;=\; \frac{f^{(N)}(z_0)}{g^{(N)}(z_0)} .$$

11. Suppose $f(z) = \sum a_k z^k$ is analytic for $|z| < R$, and suppose that $f(z)$ extends to be meromorphic for $|z| < R + \varepsilon$, with only one pole z_0 on the circle $|z| = R$. Show that $a_k/a_{k+1} \to z_0$ as $k \to \infty$.

12. Show that if z_0 is an isolated singularity of $f(z)$ that is not removable, then z_0 is an essential singularity for $e^{f(z)}$.

13. Let S be a sequence converging to a point $z_0 \in \mathbb{C}$, and let $f(z)$ be analytic on some disk centered at z_0 except possibly at the points of S and at z_0. Show that either $f(z)$ extends to be meromorphic on some neighborhood of z_0, or else for any complex number L there is a sequence $\{w_j\}$ such that $w_j \to z_0$ and $f(w_j) \to L$.

14. Suppose $u\left(re^{i\theta}\right)$ is harmonic on the punctured disk $\{0 < r < 1\}$, with Laurent series as in Exercise 7 of Section 1. Suppose $\alpha > 0$ is such that $r^\alpha u(re^{i\theta}) \to 0$ as $r \to 0$. Show that $a_n = 0 = b_n$ for $n \leq -\alpha$.

15. Suppose $u(z)$ is harmonic on the punctured disk $\{0 < |z| < \rho\}$. Show that if
$$\frac{u(|z|)}{\log(1/|z|)} \to 0$$
as $z \to 0$, then $u(z)$ extends to be harmonic at 0. What can you say if you know only that $|u(z)| \le C\log(1/|z|)$ for some fixed constant C and $0 < |z| < \rho$?

3. Isolated Singularity at Infinity

We say that $f(z)$ has an **isolated singularity at** ∞ if $f(z)$ is analytic outside some bounded set, that is, if there is $R > 0$ such that $f(z)$ is analytic for $|z| > R$. Thus $f(z)$ has an isolated singularity at ∞ if and only if $g(w) = f(1/w)$ has an isolated singularity at $w = 0$. We classify the isolated singularity of $f(z)$ at ∞ according to the isolated singularity of $g(w)$ at $w = 0$. Suppose that $f(z)$ has a Laurent series expansion
$$f(z) \;=\; \sum_{k=-\infty}^{\infty} b_k z^k, \qquad |z| > R.$$
The singularity of $f(z)$ at ∞ is **removable** if $b_k = 0$ for all $k > 0$, in which case $f(z)$ is analytic at ∞. The singularity of $f(z)$ at ∞ is **essential** if $b_k \neq 0$ for infinitely many $k > 0$. For fixed $N \ge 1$, $f(z)$ has a **pole** of order N at ∞ if $b_N \neq 0$ while $b_k = 0$ for $k > N$.

Suppose $f(z)$ has a pole of order N at ∞. The Laurent series expansion of $f(z)$ becomes
$$f(z) \;=\; b_N z^N + b_{N-1}z^{N-1} + \cdots + b_1 z + b_0 + \frac{b_{-1}}{z} + \cdots, \qquad |z| > R,$$
where $b_N \neq 0$. We define the **principal part of** $f(z)$ **at** ∞ to be the polynomial
$$P(z) \;=\; b_N z^N + b_{N-1}z^{N-1} + \cdots + b_1 z + b_0 \,.$$
The inclusion of the constant b_0 in the principal part is a matter of convenience. It guarantees that $f(z) - P(z)$ is not only analytic at ∞ but also vanishes there.

Example. Any polynomial of degree $N \ge 1$ has a pole of order N at ∞. The principal part of the polynomial coincides with the polynomial itself.

Example. The function $e^z = 1+z+z^2/2!+\cdots$ has an essential singularity at ∞.

Exercises for VI.3

1. Consider the functions in Exercise 1 of Section 2 above. Determine which have isolated singularities at ∞, and classify them.

2. Suppose that $f(z)$ is an entire function that is not a polynomial. What kind of singularity can $f(z)$ have at ∞?

3. Show that if $f(z)$ is a nonconstant entire function, then $e^{f(z)}$ has an essential singularity at $z = \infty$.

4. Show that each branch of the following functions is meromorphic at ∞, and obtain the series expansion for each branch at ∞.

   (a) $(z^2 - 1)^{5/2}$ (b) $\sqrt[3]{(z^3 - 1)}$ (c) $\sqrt{z^2 - \frac{1}{z}}$

4. Partial Fractions Decomposition

Proceeding in analogy with our earlier definition, we say that a function $f(z)$ is **meromorphic** on a domain D in the extended complex plane $\mathbb{C}^*$ if $f(z)$ is analytic on D except possibly at isolated singularities, each of which is a pole. Again, sums and products of meromorphic functions are meromorphic. Quotients of meromorphic functions are meromorphic, provided that the denominator is not identically zero.

Any rational function is meromorphic on the extended complex plane $\mathbb{C}^*$, including at ∞. We aim to establish the converse.

Theorem. *A meromorphic function on the extended complex plane $\mathbb{C}^*$ is rational.*

To see this, note first that a meromorphic function $f(z)$ on the extended complex plane can have only a finite number of poles. Otherwise, they would accumulate at a point that would not be an isolated singularity of $f(z)$. If $f(z)$ is analytic at ∞, we define $P_\infty(z)$ to be the constant function $f(\infty)$. Otherwise, $f(z)$ has a pole at ∞ and we define $P_\infty(z)$ to be the principal part of $f(z)$ at ∞. In any event, $P_\infty(z)$ is a polynomial, and $f(z) - P_\infty(z) \to 0$ as $z \to \infty$. Let $z_1, \dots, z_m$ be the poles of $f(z)$ in the finite complex plane $\mathbb{C}$, and let $P_k(z)$ be the principal part of $f(z)$ at z_k. It has the form

$$P_k(z) = \frac{\alpha_1}{z - z_k} + \frac{\alpha_2}{(z - z_k)^2} + \cdots + \frac{\alpha_n}{(z - z_k)^n},$$

and in particular, $P_k(z)$ is analytic at ∞ and vanishes there. Consider the function

$$g(z) = f(z) - P_\infty(z) - \sum_{j=1}^{m} P_j(z).$$

Since $f(z) - P_k(z)$ is analytic at z_k, and each $P_j(z)$ is analytic at z_k for $j \neq k$, $g(z)$ is analytic at each z_k. Hence $g(z)$ is an entire function, and further, $g(z) \to 0$ as $z \to \infty$. By Liouville's theorem, $g(z)$ is identically zero. Thus

$$f(z) \;=\; P_\infty(z) + \sum_{j=1}^{m} P_j(z), \tag{4.1}$$

which shows in particular that $f(z)$ is a rational function.

The decomposition (4.1) is called the **partial fractions decomposition** of the rational function $f(z)$. As a byproduct of the proof we obtain the following.

Theorem. *Every rational function has a partial fractions decomposition, expressing it as the sum of a polynomial in z and its principal parts at each of its poles in the finite complex plane.*

Suppose that $p(z)$ and $q(z)$ are polynomials. If the degree of $q(z)$ is strictly less than the degree of $p(z)$, then $p(z)/q(z) \to 0$ as $z \to \infty$. Thus $f(z) = p(z)/q(z)$ is analytic at ∞ and vanishes there, and the principal part $P_\infty(z)$ is zero. Formula (4.1) expresses $f(z)$ as the sum of the principal parts at each of its finite poles.

Example. The function $1/(z^2-1)$ is analytic at ∞ and vanishes there, and it has poles at ± 1. The partial fractions decomposition is

$$\frac{1}{z^2-1} \;=\; \frac{1}{2}\frac{1}{z-1} \;-\; \frac{1}{2}\frac{1}{z+1},$$

and these summands are the principal parts at $+1$ and -1, respectively.

For arbitrary polynomials $p(z)$ and $q(z)$, we can use the division algorithm to find the principal part $P_\infty(z)$ of $p(z)/q(z)$ at ∞. The division algorithm is a procedure that produces in a finite number of steps polynomials $P_\infty(z)$ and $r(z)$ that satisfy

$$p(z) = P_\infty(z)q(z) + r(z), \qquad \deg r(z) \;<\; \deg q(z). \tag{4.2}$$

It proceeds as follows. We assume that $q(z)$ is a monic polynomial of degree m, so that $q(z) \;=\; z^m + \cdots$. We start with a polynomial $p(z)$ of degree $n \geq m$, say $p(z) = c_0 z^n + \cdots$. For the first step we kill the top coefficient of $p(z)$ by defining $p_1(z) = p(z) - c_0 z^{n-m} q(z)$. Thus $p_1(z)$ has degree $n_1 < n$, say $p_1(z) \;=\; c_1 z^{n_1} q(z) + \cdots$. Then we repeat the first step and define $p_2(z) = p_1(z) - c_1 z^{n_1-m} q(z)$, which has degree $n_2 < n_1$. We proceed in this fashion until we reach a polynomial $p_k(z) = p_{k-1}(z) - c_{k-1} z^{n_{k-1}-m} q(z)$ such that the degree of $p_k(z)$ is less than m. This occurs after at most

$n - m + 1$ steps, and then we have

$$\begin{aligned} p(z) &= c_0 z^{n-m} q(z) + p_1(z) = c_0 z^{n-m} q(z) + c_1 z^{n_1 - m} q(z) + p_2(z) = \cdots \\ &= c_0 z^{n-m} q(z) + c_1 z^{n_1-m} q(z) + \cdots + c_{k-1} z^{n_{k-1}-m} q(z) + p_k(z). \end{aligned}$$

We set $r(z) = p_k(z)$, and we let $P_\infty(z)$ be the sum of the terms multiplying $q(z)$, and we obtain the decomposition (4.2).

Now the function $p(z)/q(z) - P_\infty(z) = r(z)/q(z)$ is analytic at ∞ and tends to 0 there. Thus the polynomial $P_\infty(z)$ coincides with the principal part of $p(z)/q(z)$ at ∞. Our recipe for obtaining the partial fractions decomposition of an arbitrary rational function $p(z)/q(z)$ is then first to obtain the polynomials $P_\infty(z)$ and $r(z)$ that satisfy (4.2) from the division algorithm, and then to find the principal parts of $r(z)/q(z)$ at each of the zeros of $q(z)$.

Example. To obtain the partial fractions decomposition of $z^3/(z^2+1)$, first express z^3 in the form given by the division algorithm (4.2), which is

$$z^3 = z(z^2+1) - z,$$

corresponding to $P_\infty(z) = z$ and $r(z) = -z$. Thus

$$\frac{z^3}{z^2+1} = z - \frac{z}{z^2+1}.$$

Then observe that the poles at $\pm i$ are simple poles, so that

$$\frac{z}{z^2+1} = \frac{\alpha}{z-i} + \frac{\beta}{z+i}$$

for some constants α and β. We put this expression over a common denominator and solve for α and β. This leads to the partial fractions decomposition

$$\frac{z^3}{z^2+1} = z - \frac{1}{2}\frac{1}{z-i} - \frac{1}{2}\frac{1}{z+i}.$$

Exercises for VI.4

1. Find the partial fractions decompositions of the following functions.

   (a) $\dfrac{1}{z^2-z}$ (c) $\dfrac{1}{(z+1)(z^2+2z+2)}$ (e) $\dfrac{z-1}{z+1}$

   (b) $\dfrac{z^2+1}{z(z^2-1)}$ (d) $\dfrac{1}{(z^2+1)^2}$ (f) $\dfrac{z^2-4z+3}{z^2-z-6}$

2. Use the division algorithm to obtain the partial fractions decomposition of the following functions.

   (a) $\dfrac{z^3+1}{z^2+1}$ (b) $\dfrac{z^9+1}{z^6-1}$ (c) $\dfrac{z^6}{(z^2+1)(z-1)^2}$

3. Let V be the complex vector space of functions that are analytic on the extended complex plane except possibly at the points 0 and i, where they have poles of order at most two. What is the dimension of V? Write down explicitly a vector space basis for V.

5. Periodic Functions

A complex number ω is a **period** of a function $f(z)$ if $f(z+\omega) = f(z)$ wherever defined. The function $f(z)$ is **periodic** if it has a period $\omega \neq 0$.

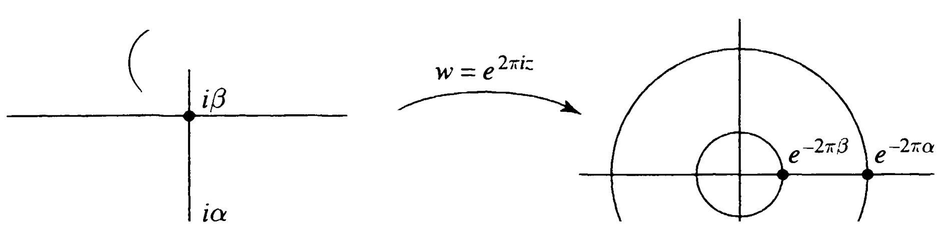
The exponential function e^z is periodic with periods $0, \pm 2\pi i, \pm 4\pi i, \ldots$. The exponential function e^{iz} is periodic with periods $0, \pm 2\pi, \pm 4\pi, \ldots$. Sums of exponential functions with the same periods are periodic. Thus $\cos z = (e^{iz} + e^{-iz})/2$ and $\sin z = (e^{iz} - e^{-iz})/2i$ are periodic. One of our goals in this section is to show that any periodic analytic function in a half-plane or strip can be represented as a sum of exponential functions.

If $\omega \neq 0$ is a period of $f(z)$, the function $g(z) = f(\omega z)$ satisfies $g(z+1) = f(\omega z + \omega) = f(\omega z) = g(z)$, so $g(z)$ has period 1. Thus we can always make a change of variable to arrange that one of the periods of a given periodic function is $\omega = 1$. We focus on functions that are analytic on a horizontal strip and that are periodic with period 1, that is, that satisfy $f(z+1) = f(z)$. This includes the exponentials $e^{2\pi ikz}$ for k an integer, and any linear combination of these exponentials.

Theorem. *If $f(z)$ is analytic on the horizontal strip $\{\alpha < \mathrm{Im}(z) < \beta\}$, and $f(z)$ is periodic with period 1, then $f(z)$ can be expanded in an absolutely convergent series of exponentials*

$$f(z) = \sum_{k=-\infty}^{\infty} a_k e^{2\pi ikz}, \qquad \alpha < \mathrm{Im}(z) < \beta.$$

The series converges uniformly on any smaller strip $\{\alpha_0 \leq \mathrm{Im}(z) \leq \beta_0\}$, where $\alpha < \alpha_0 < \beta_0 < \beta$.



To see this, we make an exponential change of variable

$$w = e^{2\pi i z}, \qquad z = -\frac{i}{2\pi}\log|w| + \frac{\arg w}{2\pi},$$

and we set $g(w) = f(z)$ with z as above. Since $f(z)$ is periodic with period 1, the value $g(w)$ does not depend on the choice of the argument of w. Thus $g(w)$ is well-defined, and $g(w)$ is analytic for $e^{-2\pi\beta} < |w| < e^{-2\pi\alpha}$. We expand $g(w)$ as a Laurent series $\sum a_k w^k$ in this annulus, and this yields the exponential series for $f(z)$.

Theorem. *Suppose $f(z)$ is analytic on the half-plane $\{\mathrm{Im}(z) > \alpha\}$, and $f(z)$ is periodic with period 1. If $f(z)$ is bounded as $\mathrm{Im}(z) \to +\infty$, then $f(z)$ can be expanded in an absolutely convergent series of exponentials*

$$f(z) = \sum_{k=0}^{\infty} a_k e^{2\pi i k z}, \qquad \mathrm{Im}(z) > \alpha.$$

The series converges uniformly on any smaller half-plane $\{\mathrm{Im}(z) \geq \alpha_0\}$, where $\alpha_0 > \alpha$.

In this case the change of variable $w = e^{2\pi i z}$ converts $f(z)$ to an analytic function $g(w)$ on the punctured disk $0 < |w| < e^{-2\pi\alpha}$. The hypothesis on $f(z)$ implies that $g(w)$ is bounded as $w \to 0$. By Riemann's theorem on removable singularities, $g(w)$ extends to be analytic at 0. Hence $g(w)$ has a power series expansion

$$g(w) = \sum_{k=0}^{\infty} a_k w^k, \qquad |w| < e^{-2\pi\alpha},$$

and this yields the exponential series for $f(z)$.

Example. The meromorphic function $1/\sin(2\pi z)$ is analytic in the upper half-plane and has period 1. From

$$|\sin(2\pi z)|^2 = \sin^2(2\pi x) + \sinh^2(2\pi y), \qquad z = x + iy,$$

we see that $1/\sin(2\pi z) \to 0$ as $y = \mathrm{Im}(z) \to +\infty$. By the preceding theorem, $1/\sin(2\pi z)$ can be expanded in a series of exponentials $e^{2\pi i k z}$ that converges absolutely in the upper half-plane. The expansion can be obtained directly from a geometric series,

$$\frac{1}{\sin(2\pi z)} = \frac{-2ie^{2\pi i z}}{1 - e^{4\pi i z}} = -2i\left[e^{2\pi i z} + e^{6\pi i z} + e^{10\pi i z} + \cdots\right].$$

Now we change our point of view. We fix a function $f(z)$, say $f(z)$ is a meromorphic function on the complex plane, and we study the set of periods of $f(z)$. If $f(z)$ is constant, then every complex number is a period of $f(z)$, and there is not much to say. So we assume that $f(z)$ is nonconstant.

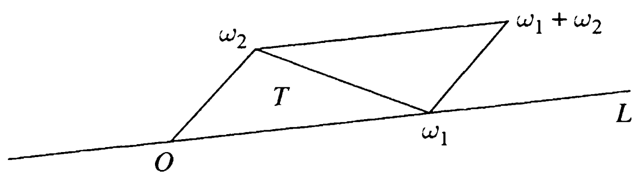
If ω_1 and ω_2 are periods of $f(z)$, then so are $m\omega_1 + n\omega_2$ for all integers m and n. (The periods form an "additive subgroup" of the complex numbers.)

Let z_0 be any point at which $f(z)$ is analytic. Since the zeros of a nonconstant analytic function are isolated, $f(z_0 + \zeta) - f(z_0) \neq 0$ on some punctured disk $\{0 < |\zeta| < \rho\}$. Since $f(z_0+\omega) - f(z_0) = 0$ for any period ω of $f(z)$, there can be no periods of $f(z)$ in the punctured disk. Thus any nonzero period ω of $f(z)$ satisfies $|\omega| \geq \rho$.

If ω and ω_2 are two different periods of $f(z)$, then $\omega_1 - \omega_2$ is a nonzero period, so $|\omega_1 - \omega_2| \geq \rho$. Since any bounded subset of the complex plane can be covered by a finite number of disks of radius $\rho/2$, and each of these contains at most one period, we conclude that any bounded subset of the complex plane contains only finitely many periods. (The periods form a "discrete subgroup" of the complex numbers.)

Let L be a straight line through 0 that contains a nonzero period, and let ω_1 be a nonzero period on L that is closest to 0. Then there cannot be a period between two consecutive integral multiples $k\omega_1$ and $(k+1)\omega_1$ of ω_1, or by subtracting we would obtain a period on L closer to 0 than ω_1. Thus the periods on L are precisely the integral multiples of ω_1.

It may occur that all the periods of $f(z)$ lie on the same straight line through 0. Otherwise, from among the periods not on the line L, choose a period ω_2 that is closest to the line segment $[0, \omega_1]$. We claim that all periods of $f(z)$ have the form $m\omega_1 + n\omega_2$ for integers m and n. We argue as follows.



Let P be the closed parallelogram with vertices 0, ω_1, ω_2, and $\omega_1 + \omega_2$. Thus P consists of precisely the complex numbers of the form $s\omega_1 + t\omega_2$ where $0 \leq s, t \leq 1$. Every complex number can be expressed in the form $m\omega_1 + n\omega_2 + z$, where m, n are integers and $z \in P$. Geometrically, this means that the parallelograms $P + m\omega_1 + n\omega_2$ fill out the complex plane. We cut P into two triangles, the triangle with vertices $0, \omega_1, \omega_2$, which we denote by T, and the triangle $\omega_1 + \omega_2 - T$ with vertices ω_1, ω_2, and $\omega_1 + \omega_2$. The only periods in T are the vertices, since any other point of T either lies on L closer to 0 than ω_1 or lies off L closer to $[0, \omega_1]$ than ω_2. It follows then that the only periods in P are the four vertices, since if ω is a period in P, then either ω or $\omega_1 + \omega_2 - \omega$ is a period belonging to T, hence a vertex of T. Since any period can be expressed in the form $m\omega_1 + n\omega_2 + \omega$, where m, n are integers and ω is a period in P, in fact every period is an integral combination of ω_1 and ω_2, as asserted.

We summarize our results as follows.

Theorem. *Suppose that $f(z)$ is a nonconstant meromorphic function on the complex plane that is periodic. Either there is a period ω_1 for $f(z)$ such that the periods of $f(z)$ are the integral multiples $m\omega_1$, $-\infty < m < \infty$, or there are two periods ω_1 and ω_2 for $f(z)$ that do not lie on the same line through the origin such that the periods of $f(z)$ are the integral combinations $m\omega_1 + n\omega_2$, $-\infty < m, n < \infty$.*

In the case that the periods of $f(z)$ all lie on the same straight line through the origin, we say that $f(z)$ is **simply periodic**. Otherwise, we say that $f(z)$ is **doubly periodic**. The entire functions e^z and $\sin z$ are simply periodic. It is possible to construct (as in Exercise 7) meromorphic functions on the complex plane that are doubly periodic. However, the only entire functions that are doubly periodic are the trivial ones.

Theorem. *An entire function that is doubly periodic is constant.*

Indeed, if the entire function $f(z)$ is doubly periodic, and if $|f(z)| \leq M$ on the parallelogram P constructed above, then by periodicity $|f(z)| \leq M$ on each translate $m\omega_1 + n\omega_2 + P$ of P. Since these translates fill out the complex plane, $f(z)$ is a bounded entire function. By Liouville's theorem, $f(z)$ is constant.

Exercises for VI.5

1. Show that if $f(z)$ and $g(z)$ have period ω, then so do $f(z) + g(z)$ and $f(z)g(z)$.

2. Expand $1/\cos(2\pi z)$ in a series of powers of $e^{2\pi iz}$ that converges in the upper half-plane. Determine where the series converges absolutely and where it converges uniformly.

3. Expand $\tan z$ in a series of powers of exponentials e^{ikz}, $-\infty < k < \infty$, that converges in the upper half-plane. Also find an expansion of $\tan z$ as an exponential series that converges in the lower half-plane.

4. Let $f(z)$ be an analytic function in the upper half-plane that is periodic, with real period $2\pi\lambda > 0$. Suppose that there are $A, C > 0$ such that $|f(x + iy)| \leq Ce^{Ay}$ for $y > 0$. Show that
$$f(z) = \sum_{n \geq -A\lambda} a_n e^{inz/\lambda},$$
where the series converges uniformly in each half-plane $\{y \geq \varepsilon\}$, for fixed $\varepsilon > 0$.

5. Suppose that ± 1 are periods of a nonzero doubly periodic function $f(z)$, and suppose that there are no periods ω of $f(z)$ satisfying $0 < |\omega| < 1$. How many periods of $f(z)$ lie on the unit circle?

Describe the possibilities, and sketch the set of periods for each possibility.

6. We say that ω_1 and ω_2 **generate** the periods of a doubly periodic function if the periods of the function are precisely the complex numbers of the form $m\omega_1 + n\omega_2$ where m and n are integers. Show that if ω_1 and ω_2 generate the periods of a doubly periodic function $f(z)$, and if λ_1 and λ_2 are complex numbers, then λ_1 and λ_2 generate the periods of $f(z)$ if and only if there is a 2×2 matrix A with integer entries and with determinant ± 1 such that $A(\omega_1, \omega_2) = (\lambda_1, \lambda_2)$.

7. Let ω_1 and ω_2 be two complex numbers that do not lie on the same line through 0. Let $k \geq 3$. Show that the series

$$\sum_{m,n=-\infty}^{\infty} \frac{1}{(z - (m\omega_1 + n\omega_2))^k}$$

converges uniformly on any bounded subset of the complex plane to a doubly periodic meromorphic function $f(z)$, whose periods are generated by ω_1 and ω_2. *Strategy.* Show that the number of periods in any annulus $\{N \leq |z| \leq N+1\}$ is bounded by CN for some constant C.

6. Fourier Series

A **complex Fourier series** is a two-tailed series of the form

$$(6.1)\quad \sum_{k=-\infty}^{\infty} c_k e^{ik\theta} \;=\; \cdots + c_{-2}e^{-2i\theta} + c_{-1}e^{-i\theta} + c_0 + c_1 e^{i\theta} + c_2 e^{2i\theta} + \cdots .$$

Laurent expansions are intimately related to Fourier series. If the Laurent series

$$f(z) \;=\; \sum_{k=-\infty}^{\infty} a_k z^k$$

converges uniformly on the circle $\{|z| = r\}$, then

$$f(re^{i\theta}) \;=\; \sum_{k=-\infty}^{\infty} a_k r^k e^{ik\theta}$$

is the Fourier series expansion of $f(re^{i\theta})$, regarded as a function of θ. The Fourier coefficients of the expansion are the coefficients $c_k = a_k r^k$.

Suppose the series (6.1) converges uniformly to a function $f\left(e^{i\theta}\right)$,

$$f(e^{i\theta}) \;=\; \sum_{j=-\infty}^{\infty} c_j e^{ij\theta}.$$

We can capture the coefficients of the series by multiplying by the exponential function $e^{-ik\theta}$ and integrating with respect to the probability measure $d\theta/2\pi$. The orthogonality relations for the exponential functions,

$$\int_{-\pi}^{\pi} e^{ij\theta} e^{-ik\theta} \frac{d\theta}{2\pi} = \begin{cases} 1, & j = k, \\ 0, & j \neq k, \end{cases} \tag{6.2}$$

then yield

$$\int_{-\pi}^{\pi} f\left(re^{i\theta}\right) e^{-ik\theta} \frac{d\theta}{2\pi} = \sum_{j=-\infty}^{\infty} c_j \int_{-\pi}^{\pi} e^{ij\theta} e^{-ik\theta} \frac{d\theta}{2\pi} = c_k.$$

This leads us to define the **Fourier coefficients** of any piecewise continuous function (or any integrable function) $f\left(e^{i\theta}\right)$ to be

$$c_k = \int_{-\pi}^{\pi} f\left(e^{i\theta}\right) e^{-ik\theta} \frac{d\theta}{2\pi}, \qquad -\infty < k < \infty, \tag{6.3}$$

and we associate to $f\left(e^{i\theta}\right)$ the Fourier series

$$f\left(e^{i\theta}\right) \sim \sum_{k=-\infty}^{\infty} c_k e^{ik\theta}, \tag{6.4}$$

where the c_k's are defined by (6.3). We call $\sum c_k e^{ik\theta}$ the **Fourier series of** $f\left(e^{i\theta}\right)$. However, we face now two big problems. Does the Fourier series of $f\left(e^{i\theta}\right)$ converge? And if so, to what?

Example. Define $f\left(e^{i\theta}\right)$ to be -1 for $-\pi < \theta < 0$, and $+1$ for $0 < \theta < \pi$. The Fourier coefficient c_0 is the average value of $f\left(e^{i\theta}\right)$, which is 0. If $k \neq 0$, then

$$\begin{aligned} c_k &= -\int_{-\pi}^{0} e^{-ik\theta} \frac{d\theta}{2\pi} + \int_{0}^{\pi} e^{-ik\theta} \frac{d\theta}{2\pi} = \left.\frac{e^{-ik\theta}}{2\pi ik}\right|_{-\pi}^{0} - \left.\frac{e^{-ik\theta}}{2\pi ik}\right|_{0}^{\pi} \\ &= \frac{1}{2\pi ik}\left[1-(-1)^k-(-1)^k+1\right] = \frac{1}{\pi ik}\left[1-(-1)^k\right]. \end{aligned}$$

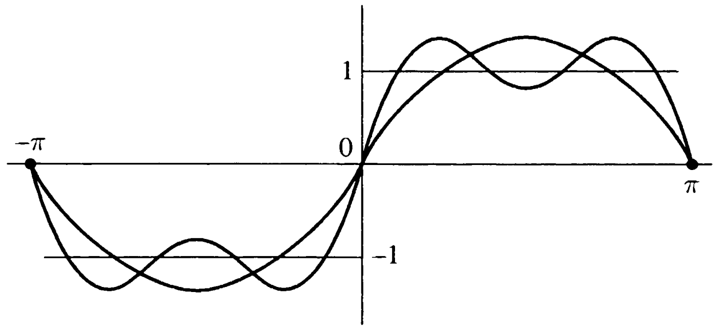
This is 0 if k is even and $2/\pi ik$ if k is odd. Thus the complex Fourier series of $f\left(e^{i\theta}\right)$ is

$$\begin{aligned} f\left(e^{i\theta}\right) \sim \frac{2}{\pi i} \sum_{k \,\mathrm{odd}} \frac{1}{k} e^{ik\theta} &= \cdots + \left(\frac{-2}{3\pi i}\right) e^{-3i\theta} + \left(\frac{-2}{\pi i}\right) e^{-i\theta} \\ &\quad + \left(\frac{2}{\pi i}\right) e^{i\theta} + \left(\frac{2}{3\pi i}\right) e^{3i\theta} + \cdots . \end{aligned}$$

If we combine the terms for $\pm k$, we obtain sine functions, and the series becomes

$$f\left(e^{i\theta}\right) \sim \frac{4}{\pi}\left(\sin\theta + \frac{1}{3}\sin 3\theta + \frac{1}{5}\sin 5\theta + \cdots\right). \tag{6.5}$$

The terms of the series are all zero at $\theta = 0$ and at $\theta = \pm\pi$. We will soon see that the series converges to $f\left(e^{i\theta}\right)$ for $0 < |\theta| < \pi$.



Theorem. *If $f\left(e^{i\theta}\right)$ is piecewise continuous (or more generally, square-integrable), with Fourier series $f\left(e^{i\theta}\right) \sim \sum c_k e^{ik\theta}$, then for $m, n \geq 0$ we have*

$$(6.6) \quad \sum_{k=-m}^{n} |c_k|^2 + \int_{-\pi}^{\pi} \left| f\left(e^{i\theta}\right) - \sum_{k=-m}^{n} c_k e^{ik\theta} \right|^2 \frac{d\theta}{2\pi} = \int_{-\pi}^{\pi} \left| f\left(e^{i\theta}\right)\right|^2 \frac{d\theta}{2\pi}.$$

This is established by writing

$$\left| f\left(e^{i\theta}\right) - \sum_{k=-m}^{n} c_k e^{ik\theta} \right|^2 = \left(f\left(e^{i\theta}\right) - \sum_{j=-m}^{n} c_j e^{ij\theta} \right) \left(\overline{f\left(e^{i\theta}\right)} - \sum_{k=-m}^{n} \overline{c_k} e^{-ik\theta} \right),$$

multiplying out the product on the right, and integrating both sides from $-\pi$ to π. The integrals featuring the products of exponentials for $j \neq k$ drop out, on account of the orthogonality relations (6.2). What remains on the right after we integrate is

$$\int_{-\pi}^{\pi} \left| f\left(e^{i\theta}\right)\right|^2 \frac{d\theta}{2\pi} - \sum \overline{c_k} \int_{-\pi}^{\pi} f\left(e^{i\theta}\right) e^{-ik\theta} \frac{d\theta}{2\pi}$$
$$- \sum c_j \int_{-\pi}^{\pi} \overline{f\left(e^{i\theta}\right)} e^{ij\theta} \frac{d\theta}{2\pi} + \sum |c_j|^2.$$

In view of formula (6.3), we recognize the second and third integrals appearing here as c_k and $\overline{c_j}$ respectively. Thus the sum is equal to

$$\int_{-\pi}^{\pi} \left| f\left(e^{i\theta}\right)\right|^2 \frac{d\theta}{2\pi} - \sum |c_k|^2 - \sum |c_j|^2 + \sum |c_j|^2.$$

If we cancel the two sums over j and move $\sum |c_k|^2$ to the other side, we obtain (6.6).

The identity (6.6) shows that the partial sums of the series $\sum |c_k|^2$ are bounded. Consequently, the series $\sum |c_k|^2$ converges, and from (6.6) we obtain the following estimate.

Theorem (Bessel's Inequality). *If $f(e^{i\theta})$ is piecewise continuous (or more generally, square-integrable), with Fourier series $f(e^{i\theta}) \sim \sum c_k e^{ik\theta}$, then*

$$\sum_{k=-\infty}^{\infty} |c_k|^2 \leq \int_{-\pi}^{\pi} \left|f\left(e^{i\theta}\right)\right|^2 \frac{d\theta}{2\pi}. \tag{6.7}$$

From Bessel's inequality it follows in particular that $c_k \to 0$ as $k \to \pm\infty$. With this observation in hand, we are now ready to state and prove our main theorem on pointwise convergence of Fourier series.

Theorem. *Suppose $f(e^{i\theta})$ is piecewise continuous (or square integrable), with Fourier series $f(e^{i\theta}) \sim \sum c_k e^{ik\theta}$. If $f(e^{i\theta})$ is differentiable at θ_0, then the Fourier series of $f(e^{i\theta})$ converges to $f(e^{i\theta_0})$ at $\theta = \theta_0$,*

$$f(e^{i\theta_0}) = \sum_{-\infty}^{\infty} c_k e^{ik\theta_0} = \lim_{m,n\to\infty} \sum_{k=-m}^{n} c_k e^{ik\theta_0}.$$

Though Fourier series had been studied intensively for well over a century, this ingenious proof was discovered only relatively recently, by P. Chernoff in 1980.

We consider first the special case in which $\theta_0 = 0$ and $e^{i\theta_0} = 1$. Define $g(e^{i\theta}) = \left[f(e^{i\theta}) - f(1)\right] / \left(e^{i\theta} - 1\right)$. The differentiability of $f(e^{i\theta})$ at $\theta = 0$ implies that $g(e^{i\theta})$ has a limit as $\theta \to 0$. Consequently, $g(e^{i\theta})$ is also piecewise continuous. Denote the Fourier coefficients of $g(e^{i\theta})$ by b_k, so that $g(e^{i\theta}) \sim \sum b_k e^{ik\theta}$. Bessel's inequality for $g(e^{i\theta})$ shows that $b_k \to 0$ as $k \to \pm\infty$. Now we compute the c_k's in terms of the b_k's. Since $f(e^{i\theta}) = g(e^{i\theta})(e^{i\theta} - 1) + f(1)$, we have

$$c_k = \int_{-\pi}^{\pi} g\left(e^{i\theta}\right)\left(e^{i\theta} - 1\right)e^{-ik\theta}\frac{d\theta}{2\pi} + f(1)\int_{-\pi}^{\pi} e^{-ik\theta}\frac{d\theta}{2\pi}.$$

When we express these integrals as Fourier coefficients of $g(e^{i\theta})$, we obtain $c_k = b_{k-1} - b_k$ if $k \neq 0$, and $c_0 = b_{-1} - b_0 + f(1)$. Hence the series $\sum c_k$ telescopes, and we obtain

$$\sum_{k=-m}^{n} c_k = f(1) + \sum_{k=-m}^{n} (b_{k-1} - b_k) = f(1) + b_{-m-1} - b_n,$$

which tends to $f(1)$ as $m, n \to +\infty$. This proves the theorem when $\theta_0 = 0$.

The case when θ_0 is arbitrary is reduced to the above special case by a change of variable. Consider the function $h(e^{i\theta}) = f\left(e^{i(\theta+\theta_0)}\right) \sim \sum a_k e^{ik\theta}$, which is piecewise continuous and which is differentiable at $\theta = 0$.

The Fourier coefficient a_k of $h\left(e^{i\theta}\right)$ is

$$a_k = \int_{-\pi}^{\pi} f\left(e^{i(\theta+\theta_0)}\right) e^{-ik\theta}\frac{d\theta}{2\pi} = \int_{-\pi}^{\pi} f\left(e^{ik\varphi}\right) e^{-ik\varphi}e^{ik\theta_0}\frac{d\varphi}{2\pi} = c_k e^{ik\theta_0}.$$

Thus the Fourier series of $f\left(e^{i\theta}\right)$ evaluated at $\theta=\theta_0$ is $\sum c_k e^{ik\theta_0} = \sum a_k$, which is the same as the Fourier series of $h\left(e^{i\theta}\right)$ evaluated at $\theta = 0$. We have shown that the latter converges to $h(1) = f(e^{i\theta_0})$. This completes the proof.

Example. Consider the series in (6.5). By the convergence theorem, the series converges to $+1$ for $0<\theta<\pi$ and to -1 for $-\pi<\theta<0$. The convergence theorem does not give any information about the points $\theta = 0$ and $\theta=\pm\pi$, where the function is discontinuous. However, we see directly that the series converges to 0 at these points.

We aim now to establish a result on uniform convergence of Fourier series. We begin by showing that the Fourier series of a smooth function can be differentiated term by term.

Theorem. *Suppose $f\left(e^{i\theta}\right)$ is a continuously differentiable function of θ, with Fourier series $f\left(e^{i\theta}\right) \sim \sum c_k e^{ik\theta}$. Then the Fourier series of the derivative of $f\left(e^{i\theta}\right)$ is obtained by differentiating term by term,*

$$\frac{d}{d\theta}f\left(e^{i\theta}\right) \sim \sum ikc_k e^{ik\theta}.$$

To check this, we simply write down the expression for the kth Fourier coefficient of the derivative and we integrate by parts. The Fourier coefficient of the derivative is

$$\int_{-\pi}^{\pi} e^{-ik\theta}\frac{d}{d\theta}f\left(e^{i\theta}\right)\frac{d\theta}{2\pi} = e^{-ik\theta}f\left(e^{i\theta}\right)\Big|_{-\pi}^{\pi} + ik\int_{-\pi}^{\pi} f\left(e^{i\theta}\right)e^{-ik\theta}\frac{d\theta}{2\pi} = ikc_k\,.$$

If we combine this theorem with Bessel's inequality, we can show that the Fourier coefficients of an n-times continuously differentiable function tend to zero at least as rapidly as the nth power of $1/k$. The smoother the function, the more rapidly its Fourier coefficients decay.

Corollary. *If $f\left(e^{i\theta}\right)$ is an n-times continuously differentiable function of θ, with Fourier series $f\left(e^{i\theta}\right) \sim \sum c_k e^{ik\theta}$, then $\sum_{k=-\infty}^{\infty} k^{2n}|c_k|^2<\infty$. Further, $k^n c_k \to 0$ as $k\to\pm\infty$.*

To see this, observe that the nth derivative of $f\left(e^{i\theta}\right)$ has Fourier series $\sum(ik)^n c_k e^{ik\theta}$ and apply Bessel's inequality. The second statement of the corollary follows immediately from the convergence of the series, since the terms of the series then tend to 0.

In particular, if $f\left(e^{i\theta}\right)$ is a twice continuously differentiable function of θ, then $k^2 c_k \to 0$ as $k \to \pm\infty$. Hence the series $\sum |c_k|$ converges, by comparison with $\sum 1/k^2$. By the Weierstrass M-test, the series $\sum c_k e^{ik\theta}$ then converges uniformly in θ. By the pointwise convergence theorem proved above, the sum of the series is $f\left(e^{i\theta}\right)$. We have proved the following.

Theorem. *Suppose $f\left(e^{i\theta}\right)$ is a twice continuously differentiable function of θ. Then the Fourier series of $f\left(e^{i\theta}\right)$ converges to $f\left(e^{i\theta}\right)$ uniformly in θ.*

Example. Define $g(\theta) = \theta^4 - 2\pi^2\theta^2$, $-\pi \le \theta \le \pi$. We check that $g(-\pi) = g(\pi)$, $g'(-\pi) = g'(\pi)$, and $g''(-\pi) = g''(\pi)$. Consequently, if we set $f\left(e^{i\theta}\right) = g(\theta)$, we obtain a twice continuously differentiable function of $e^{i\theta}$. By the theorem, the Fourier series of $f\left(e^{i\theta}\right)$ converges uniformly in θ. The explicit calculation of the Fourier series is left to the exercises (Exercise 4).

Example. The Fourier series in (6.5) does not converge uniformly, since the sum is not continuous.

Exercises for VI.6

1. Consider the continuous function $f(e^{i\theta}) = |\theta|$, $-\pi \le \theta \le \pi$. Find the complex Fourier series of $f\left(e^{i\theta}\right)$ and show that it can be expressed as a cosine series. Sketch the graphs of the first three partial sums of the cosine series. Discuss the convergence of the series. Does it converge uniformly? *Partial answer.* The cosine series is
$$|\theta| = \frac{\pi}{2} - \frac{4}{\pi}\left(\cos\theta + \frac{1}{3^2}\cos 3\theta + \frac{1}{5^2}\cos 5\theta + \cdots\right).$$

2. Let $f(e^{i\theta}) = \theta$, $-\pi < \theta \le \pi$ (the principal value of the argument). Find the complex Fourier series of $f\left(e^{i\theta}\right)$ and the sine series of $f\left(e^{i\theta}\right)$. Show that the complex Fourier series diverges at $\theta = \pm\pi$, while the sine series converges at $\pm\pi$. Differentiate the complex Fourier series term by term and determine where the differentiated series converges.

3. Consider the continuous function $f(e^{i\theta}) = \theta^2$, $-\pi \le \theta \le \pi$. Find the complex Fourier series of $f\left(e^{i\theta}\right)$ and show that it can be expressed as a cosine series. Discuss the convergence of the series. Does it converge uniformly? By substituting $\theta = 0$, show that
$$\frac{\pi^2}{12} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots.$$

4. Consider the continuous function $f(e^{i\theta}) = \theta^4 - 2\pi^2\theta^2$, $-\pi \le \theta \le \pi$. Find the complex Fourier series of $f\left(e^{i\theta}\right)$ and show that it can be

expressed as a cosine series. Relate the Fourier series to the series of the function in Exercise 3.

5. Show that if $\sum c_k$ converges absolutely, then $\sum c_k e^{ik\theta}$ converges absolutely for each θ, and the series converges uniformly for $-\pi \leq \theta \leq \pi$.

6. Show that any function $f\left(e^{i\theta}\right)$ on the unit circle with absolutely convergent Fourier series has the form $f\left(e^{i\theta}\right) = g\left(e^{i\theta}\right) + \overline{h\left(e^{i\theta}\right)}$, where $g(z)$ and $h(z)$ are continuous functions on the unit circle that extend continuously to be analytic on the open unit disk.

7. If $f\left(e^{i\theta}\right) \sim \sum c_k e^{ik\theta}$, and the series converges uniformly to $f\left(e^{i\theta}\right)$, then

$$\int_{-\pi}^{\pi} \left|f\left(e^{i\theta}\right)\right|^2 \frac{d\theta}{2\pi} = \sum_{k=-\infty}^{\infty} |c_k|^2 .$$

Remark. This is called **Parseval's identity**. Formula (6.6) shows that Parseval's identity holds for a function $f\left(e^{i\theta}\right)$ if and only if the partial sums of the Fourier series of $f\left(e^{i\theta}\right)$ converge to $f\left(e^{i\theta}\right)$ in the sense of "mean-square" or "L^2-approximation."

8. By applying Parseval's identity to the piecewise constant function with series (6.5), show that

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots .$$

Use this identity and some algebraic manipulation to show that

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots .$$

9. By applying Parseval's identity to the function of Exercise 1, show that

$$\frac{\pi^4}{96} = 1 + \frac{1}{3^4} + \frac{1}{5^4} + \frac{1}{7^4} + \cdots .$$

Use this identity and some algebraic manipulation to show that

$$\frac{\pi^4}{90} = 1 + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots .$$

10. If $f(z)$ is analytic in some annulus containing the unit circle $|z| = 1$, with Laurent expansion $\sum a_k z^k$, then

$$\frac{1}{2\pi} \oint_{|z|=1} |f(z)|^2 |dz| = \sum_{k=-\infty}^{\infty} |a_k|^2.$$

11. Let $f\left(e^{i\theta}\right)$ be a continuous function on the unit circle, with Fourier series $\sum c_k e^{ik\theta}$. Show that $f\left(e^{i\theta}\right)$ extends to be analytic on some annulus containing the unit circle if and only if there exist $r < 1$ and $C > 0$ such that $|c_k| \le Cr^{|k|}$ for $-\infty < k < \infty$.

12. Using the convergence theorem for Fourier series, prove that every continuous function on the unit circle in the complex plane can be approximated uniformly there by trigonometric polynomials, that is, by finite linear combinations of exponentials $e^{ik\theta}$, $-\infty < k < \infty$. *Strategy.* First approximate $f\left(e^{i\theta}\right)$ by a smooth function.

13. Let D be a domain bounded by a smooth boundary curve of length 2π. We parametrize the boundary of D by arc length s, so the boundary is given by a smooth periodic function $\gamma(s)$, $0 \le s \le 2\pi$. Let $\sum c_k e^{iks}$ be the Fourier series of $\gamma(s)$. (a) Show that $\sum k^2|c_k|^2 = 1$. *Hint.* Apply Parseval's identity to $\gamma'(s)$ and use $|\gamma'(s)| = 1$ for a curve parameterized by arc length. (b) Show that the area of D is $\pi \sum k|c_k|^2$. *Hint.* Use Exercise IV.1.4. (c) Show that the area of D is $\le \pi$, with equality if and only if D is a disk. *Remark.* This proves the **isoperimetric theorem**: *Among all smooth closed curves of a given length, the curve that surrounds the largest area is a circle.*

14. Show that

$$\int_{-\pi}^{\pi}\left|f\left(e^{i\theta}\right) - \sum_{k=-m}^{n} b_k e^{ik\theta}\right|^2 \frac{d\theta}{2\pi}$$
$$= \int_{-\pi}^{\pi}\left|f\left(e^{i\theta}\right) - \sum_{k=-m}^{n} c_k e^{ik\theta}\right|^2 \frac{d\theta}{2\pi} + \sum_{k=-m}^{n} |b_k - c_k|^2$$

for any choice of complex numbers b_k, $-m \le k \le n$. *Remark.* This shows that the best mean-square approximant to $f\left(e^{i\theta}\right)$ by exponential sums $\sum_{-m}^{n} b_k e^{ik\theta}$, for fixed m and n, is the corresponding partial sum of the Fourier series.

15. Show that a continuously differentiable function on the unit circle has an absolutely convergent Fourier series. *Strategy.* Write the Fourier coefficients c_k of $f\left(e^{i\theta}\right)$ as $a_k b_k$, where $a_k = 1/ik$ and b_k is the Fourier coefficient of the derivative. Use Bessel's inequality and the Cauchy-Schwarz inequality $\left|\sum \alpha_k \overline{\beta_k}\right| \le \sqrt{\sum |\alpha_k|^2}\sqrt{\sum |\beta_k|^2}$.

16. Let $f\left(e^{i\theta}\right)$ be a continuous function on the unit circle. Suppose that $f\left(e^{i\theta}\right)$ is piecewise continuously differentiable, in the sense that it has a continuous derivative except at a finite number of points, at each of which the derivative has limits from the left and from the right. Show that the Fourier series of $f\left(e^{i\theta}\right)$ is absolutely convergent. *Strategy.* Cancel the discontinuities of the derivative

using translates of the function in Exercise 3, whose Fourier series is absolutely convergent.

17. Let $f\left(e^{i\theta}\right)$ be piecewise continuously differentiable, in the sense that it is continuously differentiable except at a finite number of points, at each of which both the function and its derivative have limits from the left and from the right. Show that the Fourier series of $f\left(e^{i\theta}\right)$ converges at each point, to $f\left(e^{i\theta}\right)$ if the function is continuous at $e^{i\theta}$, and otherwise to the average of the limits of $f\left(e^{i\theta}\right)$ from the left and from the right. *Strategy.* Show that $f\left(e^{i\theta}\right) = f_1\left(e^{i\theta}\right)+\sum b_j h_j\left(e^{i\theta}\right)$, where $f_1\left(e^{i\theta}\right)$ satisfies the hypotheses of Exercise 15, and each $h_j\left(e^{i\theta}\right)$ is obtained from the function of Exercise 2 by a change of variable $\theta \mapsto \theta - \theta_j$.

VII

The Residue Calculus

Section 1 is devoted to the residue theorem and to techniques for evaluating residues. In the remaining sections we apply the residue theorem to evaluate various real integrals. This material provides a good training ground for the techniques of complex integration. The student who is anxious to move on can skip the final several sections of the chapter at first reading.

1. The Residue Theorem

Suppose z_0 is an isolated singularity of $f(z)$ and that $f(z)$ has Laurent series

$$f(z) \;=\; \sum_{n=-\infty}^{\infty} a_n(z-z_0)^n, \qquad 0 < |z-z_0| < \rho.$$

We define the **residue** of $f(z)$ at z_0 to be the coefficient a_{-1} of $1/(z-z_0)$ in this Laurent expansion,

$$\text{(1.1)} \qquad \operatorname{Res}\,[\,f(z), z_0\,] \;=\; a_{-1} \;=\; \frac{1}{2\pi i}\oint_{|z-z_0|=r} f(z)\,dz\,,$$

where r is any fixed radius satisfying $0 < r < \rho$.

Example. The definition yields immediately

$$\operatorname{Res}\left[\frac{1}{z}\,,\,0\right] \;=\; 1, \qquad \operatorname{Res}\left[\frac{1}{(z-z_0)^2}\,,\,z_0\right] \;=\; 0.$$

Example. The partial fractions decomposition

$$\frac{1}{z^2+1} \;=\; \frac{1}{2i}\left[\frac{1}{z-i} \;-\; \frac{1}{z+i}\right] \;=\; \frac{1}{2i}\,\frac{1}{z-i} \;+\; [\text{analytic at } i]$$

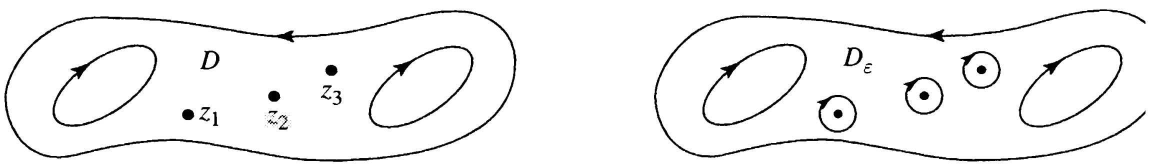
yields

$$\operatorname{Res}\left[\frac{1}{z^2+1}\,,\,i\right] \;=\; \frac{1}{2i}\,.$$

The following residue theorem provides an important tool for evaluating complex line integrals. It extends Cauchy's theorem by allowing for a finite number of singularities inside the contour of integration. When there are no singularities present, the residue theorem reduces to Cauchy's theorem.

Theorem (Residue Theorem). *Let D be a bounded domain in the complex plane with piecewise smooth boundary. Suppose that $f(z)$ is analytic on $D \cup \partial D$, except for a finite number of isolated singularities $z_1, \ldots, z_m$ in D. Then*

$$\int_{\partial D} f(z)\,dz \;=\; 2\pi i \sum_{j=1}^{m} \operatorname{Res}[f(z), z_j]. \tag{1.2}$$



To see this, let D_ε be the domain obtained from D by punching out small disks U_j centered at z_j of radius ε. The formula (1.1) for the residue at z_j yields

$$\int_{\partial U_j} f(z)\,dz \;=\; 2\pi i \operatorname{Res}[f(z), z_j].$$

(This can be regarded as a special case of the residue theorem, for the domain U_j and a function with a singularity at z_j.) By Cauchy's theorem,

$$0 \;=\; \int_{\partial D_\varepsilon} f(z)\,dz \;=\; \int_{\partial D} f(z)\,dz \;-\; \sum_{j=1}^{m} \int_{\partial U_j} f(z)\,dz.$$

If we combine these two identities, we obtain (1.2).

We give four useful rules for calculating residues.

Rule 1. *If $f(z)$ has a simple pole at z_0, then*

$$\operatorname{Res}[f(z), z_0] \;=\; \lim_{z \to z_0} (z - z_0) f(z).$$

In this case the Laurent series of $f(z)$ is

$$f(z) \;=\; \frac{a_{-1}}{z - z_0} \;+\; [\text{analytic at } z_0],$$

from which the rule follows immediately. Note that once we obtain an expression for $(z - z_0)f(z)$ as an analytic function, the limit is evaluated by simply plugging $z = z_0$ into the expression.

Example. From Rule 1 we have

$$\operatorname{Res}\left[\frac{1}{z^2+1}, i\right] = \lim_{z\to i} \frac{z-i}{z^2+1} = \lim_{z\to i} \frac{1}{z+i} = \left.\frac{1}{z+i}\right|_{z=i} = \frac{1}{2i}.$$

This method of obtaining the residue is faster than finding first the partial fractions decomposition. Rule 4 below is faster yet.

Rule 2. *If $f(z)$ has a double pole at z_0, then*

$$\operatorname{Res}[f(z), z_0] = \lim_{z\to z_0} \frac{d}{dz}[(z-z_0)^2 f(z)].$$

In this case the Laurent expansion is

$$f(z) = \frac{a_{-2}}{(z-z_0)^2} + \frac{a_{-1}}{z-z_0} + a_0 + \cdots.$$

Thus

$$(z-z_0)^2 f(z) = a_{-2} + a_{-1}(z-z_0) + a_0(z-z_0)^2 + \cdots.$$

If we differentiate and then plug in $z = z_0$, we obtain Rule 2. Note that Rule 2 can be regarded as the formula for the coefficient of $z - z_0$ in the power series expansion of the analytic function $(z-z_0)^2 f(z)$.

Example. The function $1/(z^2+1)^2$ has double poles at $\pm i$. The residue at i is given by

$$\operatorname{Res}\left[\frac{1}{(z^2+1)^2}, i\right] = \lim_{z\to i} \frac{d}{dz}\frac{1}{(z+i)^2} = \left.\frac{-2}{(z+i)^3}\right|_{z=i} = \frac{1}{4i}.$$

Rule 3. *If $f(z)$ and $g(z)$ are analytic at z_0, and if $g(z)$ has a simple zero at z_0, then*

$$\operatorname{Res}\left[\frac{f(z)}{g(z)}, z_0\right] = \frac{f(z_0)}{g'(z_0)}.$$

In this case $f(z)/g(z)$ has at most a simple pole at z_0. If we use Rule 1 and the definition of the derivative, we obtain for the residue

$$\lim_{z\to z_0} (z-z_0)\frac{f(z)}{g(z)} = \lim_{z\to z_0} \frac{f(z)}{(g(z)-g(z_0))/(z-z_0)} = \frac{f(z_0)}{g'(z_0)}.$$

Example. The partial fractions decomposition of the function $z^3/(z^2+1)$ was found in Section VI.4 to be

$$\frac{z^3}{z^2+1} = z - \frac{1}{2}\frac{1}{z-i} - \frac{1}{2}\frac{1}{z+i}.$$

From this we read off the residues at $\pm i$ to be both $-\frac{1}{2}$. The residues can also be obtained directly using Rule 3. The residue at i is given by

$$\operatorname{Res}\left[\frac{z^3}{z^2+1}, i\right] = \left.\frac{z^3}{2z}\right|_{z=i} = \frac{i^3}{2i} = -\frac{1}{2}.$$

The following special case of Rule 3 is particularly useful.

Rule 4. *If $g(z)$ is analytic and has a simple zero at z_0, then*

$$\operatorname{Res}\left[\frac{1}{g(z)}, z_0\right] = \frac{1}{g'(z_0)}.$$

Example. If we apply Rule 4 to $1/(z^2+1)$, we obtain the residue even faster than before,

$$\operatorname{Res}\left[\frac{1}{z^2+1}, i\right] = \left.\frac{1}{2z}\right|_{z=i} = \frac{1}{2i}.$$

Exercises for VII.1

1. Evaluate the following residues.

(a) $\operatorname{Res}\left[\frac{1}{z^2+4}, 2i\right]$ (d) $\operatorname{Res}\left[\frac{\sin z}{z^2}, 0\right]$ (g) $\operatorname{Res}\left[\frac{z}{\operatorname{Log} z}, 1\right]$

(b) $\operatorname{Res}\left[\frac{1}{z^2+4}, -2i\right]$ (e) $\operatorname{Res}\left[\frac{\cos z}{z^2}, 0\right]$ (h) $\operatorname{Res}\left[\frac{e^z}{z^5}, 0\right]$

(c) $\operatorname{Res}\left[\frac{1}{z^5-1}, 1\right]$ (f) $\operatorname{Res}\left[\cot z, 0\right]$ (i) $\operatorname{Res}\left[\frac{z^n+1}{z^n-1}, e^{2\pi}\right.$

2. Calculate the residue at each isolated singularity in the complex plane of the following functions.

(a) $e^{1/z}$ (b) $\tan z$ (c) $\frac{z}{(z^2+1)^2}$ (d) $\frac{1}{z^2+z}$

3. Evaluate the following integrals, using the residue theorem.

(a) $\oint_{|z|=1} \frac{\sin z}{z^2}\,dz$ (c) $\oint_{|z|=2} \frac{z}{\cos z}\,dz$ (e) $\oint_{|z-1|=1} \frac{1}{z^8-1}\,dz$

(b) $\oint_{|z|=2} \frac{e^z}{z^2-1}\,dz$ (d) $\oint_{|z|=1} \frac{z^4}{\sin z}\,dz$ (f) $\oint_{|z-1/2|=3/2} \frac{\tan z}{z}\,dz$

4. Suppose $P(z)$ and $Q(z)$ are polynomials such that the zeros of $Q(z)$ are simple zeros at the points $z_1, \dots, z_m$, and $\deg P(z) < \deg Q(z)$.

Show that the partial fractions decomposition of $P(z)/Q(z)$ is given by

$$\frac{P(z)}{Q(z)} = \sum_{j=1}^{m} \frac{P(z_j)}{Q'(z_j)} \frac{1}{z - z_j}.$$

5. Let $f(z)$ be a meromorphic function on the complex plane that is doubly periodic, and suppose that none of the poles of $f(z)$ lie on the boundary of the period parallelogram P constructed in Section VI.5. By integrating $f(z)$ around the boundary of P, show that the sum of the residues at the poles of $f(z)$ in P is zero. Conclude that there is no doubly periodic meromorphic function with only one pole, a simple pole, in the period parallelogram.

6. Consider the integral

$$\int_{\partial D_R} \frac{e^{\pi i (z-1/2)^2}}{1 - e^{-2\pi i z}}\, dz,$$

where D_R is the parallelogram with vertices $\pm(\frac{1}{2}) \pm (1+i)R$. (a) Use the residue theorem to show that the integral is $(1+i)/\sqrt{2}$. (b) By parameterizing the sides of the parallelogram, show that the integral tends to

$$(1+i)\int_{-\infty}^{\infty} e^{-2\pi t^2}\, dt$$

as $R \to \infty$. (c) Use (a) and (b) to show that

$$\int_{-\infty}^{\infty} e^{-s^2}\, ds = \sqrt{\pi}.$$

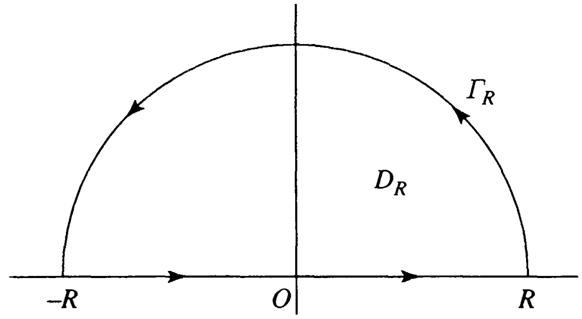
2. Integrals Featuring Rational Functions

The prototype for evaluation of an integral by means of contour integration is the derivation of the formula

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi. \tag{2.1}$$

This integral can, of course, be evaluated using the usual integration formula featuring the inverse tangent function. To evaluate it using contour integration, we proceed as follows.

Let D_R be the half-disk in the upper half-plane bounded by the interval $[-R, R]$ on the real axis and the semicircular contour Γ_R of radius R in the upper half-plane. The function $1/(1+z^2)$ has one pole in D_R, a simple



senicircular contour

pole at i with residue $1/2i$. The residue theorem yields

$$\int_{\partial D_R} \frac{dz}{1+z^2} = 2\pi i \operatorname{Res}\left[\frac{1}{1+z^2}, i\right] = 2\pi i \cdot \frac{1}{2i} = \pi.$$

Now,

$$\int_{\partial D_R} \frac{dz}{1+z^2} = \int_{-R}^{R} \frac{dx}{1+x^2} + \int_{\Gamma_R} \frac{dz}{1+z^2}.$$

On Γ_R we have

$$|1/(1+z^2)| \leq 1/(R^2-1) \sim 1/R^2,$$

while the length of Γ_R is πR. By the ML-estimate we have

$$\left| \int_{\Gamma_R} \frac{dz}{1+z^2} \right| \leq \frac{1}{R^2-1} \cdot \pi R \sim \frac{1}{R},$$

which tends to 0 as $R \to \infty$. Hence

$$\lim_{R\to\infty} \int_{-R}^{R} \frac{dx}{1+x^2} = \pi,$$

which is (2.1).

The same technique can be used to evaluate integrals of the form

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)}\, dx, \tag{2.2}$$

where $P(z)$ and $Q(z)$ are polynomials, and $Q(z)$ has no zeros on the real axis. For convergence of the integral, we require that

$$\deg Q(z) \geq \deg P(z) + 2. \tag{2.3}$$

The integral is evaluated by integrating $P(z)/Q(z)$ around the boundary of a half-disk in the upper half-plane, as above, and letting the radius tend to ∞. This yields the formula

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)}\, dx = 2\pi i \sum \operatorname{Res}\left[\frac{P(z)}{Q(z)}, z_j\right], \tag{2.4}$$

summed over the poles z_j of $P(z)/Q(z)$ in the upper half-plane.

The same contour can be used to evaluate the integrals of rational functions times trigonometric functions. The typical integral has the form

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} \cos(ax)\, dx\,,$$

where the polynomials $P(z)$ and $Q(z)$ have real coefficients and satisfy (2.3). To use the semicircular contour, we cannot use the function $\cos(az)$, because it behaves too badly in the upper half-plane. (As we have seen, the cosine function is essentially a hyperbolic cosine on the imaginary axis, which grows exponentially fast.) To obtain a tractable integral, we resort to a trick. The trick is to substitute e^{iz} for the cosine function in the contour integral, and to recover the cosine integral at the end by taking real parts. Since $|e^{i(x+iy)}| = e^{-y}$, the exponential function e^{iz} is bounded by 1 in the upper half-plane,

$$|e^{iz}| \le 1, \qquad \mathrm{Im}(z) \ge 0.$$

Example. We show by contour integration that

$$\int_{-\infty}^{\infty} \frac{\cos(ax)}{1+x^2}\, dx = \pi e^{-a}, \qquad a > 0. \tag{2.5}$$

Again we let D_R be the half-disk in the upper half-plane bounded by the interval $[-R, R]$ on the real axis and the semicircular contour Γ_R of radius R in the upper half-plane. This time we integrate the function $e^{iaz}/(1+z^2)$ over the boundary of D_R. The function has only one pole in the upper half-plane, a simple pole at i, with residue calculated by Rule 3 to be

$$\mathrm{Res}\left[\frac{e^{iaz}}{1+z^2}, i\right] = \left.\frac{e^{iaz}}{2z}\right|_{z=i} = \frac{e^{-a}}{2i}.$$

Thus

$$\int_{\partial D_R} \frac{e^{iaz}}{1+z^2}\, dz = 2\pi i \cdot \frac{e^{-a}}{2i} = \pi e^{-a}.$$

Since $|e^{iaz}| \le 1$ in the upper half-plane, the ML-estimate yields

$$\left| \int_{\Gamma_R} \frac{e^{iaz}}{1+z^2}\, dz \right| \le \frac{1}{R^2-1} \cdot \pi R \sim \frac{1}{R}.$$

Again we have

$$\int_{\partial D_R} \frac{e^{iaz}}{1+z^2}\, dz = \int_{-R}^{R} \frac{e^{iax}}{1+x^2}\, dx + \int_{\Gamma_R} \frac{e^{iaz}}{1+z^2}\, dz\,.$$

Passing to the limit as $R \to \infty$, we obtain

$$\int_{-\infty}^{\infty} \frac{e^{iax}}{1+x^2}\, dx = \pi e^{-a}, \qquad a > 0.$$

Now we take the real parts of the integral, and we obtain (2.5). Note that if we take the imaginary part of the integral, we obtain

$$\int_{-\infty}^{\infty} \frac{\sin(ax)}{1+x^2}\,dx \;=\; 0,$$

which is no surprise, since the integrand is an odd function.

Exercises for VII.2

1. Show using residue theory that

$$\int_{-\infty}^{\infty} \frac{dx}{x^2+a^2} \;=\; \frac{\pi}{a}, \qquad a>0.$$

Remark. Check the result by evaluating the integral directly, using the arctangent function.

2. Show using residue theory that

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)^2} \;=\; \frac{\pi}{2a^3}\,.$$

Remark. Check the result by differentiating the formula in the preceding exercise with respect to the parameter.

3. Show using residue theory that

$$\int_{-\infty}^{\infty} \frac{x^2\,dx}{(x^2+1)^2} \;=\; \frac{\pi}{2}\,.$$

Remark. Check the result by combining the preceding two exercises.

4. Using residue theory, show that $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{x^4+1} \;=\; \frac{\pi}{\sqrt{2}}\,.$

5. Using residue theory, show that $\displaystyle\int_{0}^{\infty} \frac{x^2}{x^4+1}\,dx \;=\; \frac{\pi}{2\sqrt{2}}\,.$

6. Show that $\displaystyle\int_{-\infty}^{\infty} \frac{x}{(x^2+2x+2)(x^2+4)}\,dx \;=\; -\frac{\pi}{10}\,.$

7. Show that

$$\int_{-\infty}^{\infty} \frac{\cos(ax)}{x^4+1}\,dx \;=\; \frac{\pi}{\sqrt{2}} e^{-a/\sqrt{2}} \left(\cos\frac{a}{\sqrt{2}} + \sin\frac{a}{\sqrt{2}} \right), \qquad a>0.$$

8. Show that

$$\int_{-\infty}^{\infty} \frac{\cos x}{(1+x^2)^2}\,dx \;=\; 2\pi e^{-1/\sqrt{2}} \left[\sqrt{2}\sin\frac{1}{\sqrt{2}} - \cos\frac{1}{\sqrt{2}} \right].$$

9. Show that

$$\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2+1}\, dx \;=\; \frac{\pi}{2}\left[1 - \frac{1}{e^2}\right].$$

10. Show that

$$\int_{-\infty}^{\infty} \frac{\cos(ax)}{x^2+b^2}\, dx \;=\; \frac{\pi e^{-|a|b}}{b}, \qquad -\infty < a < \infty,\ b > 0.$$

For which complex values of the parameters a and b does the integral exist? Where does the integral depend analytically on the parameters?

11. Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{(x^2+a^2)(x^2+b^2)}\, dx$. Indicate the range of the parameters a and b.

12. Let $Q(z)$ be a polynomial of degree m with no zeros on the real line, and let $f(z)$ be a function that is analytic in the upper half-plane and across the real line. Suppose there is $b < m - 1$ such that $|f(z)| \le |z|^b$ for z in the upper half-plane, $|z| > 1$. Show that

$$\int_{-\infty}^{\infty} \frac{f(x)}{Q(x)}\, dx \;=\; 2\pi i \sum \operatorname{Res}\left[\frac{f(z)}{Q(z)}, z_j\right],$$

summed over the zeros z_j of $Q(z)$ in the upper half-plane.

3. Integrals of Trigonometric Functions

We have seen how complex contour integrals along a curve can be converted to garden-variety integrals by parameterizing the curve. Some definite integrals can be evaluated through the reverse process, by converting them to complex contour integrals and using the residue theorem. To illustrate this, we show how the integral

$$\int_0^{2\pi} \frac{d\theta}{a+\cos\theta}, \qquad a > 1, \tag{3.1}$$

can be evaluated using residue calculus. The idea is to convert this to a complex contour integral around the unit circle. The usual parameterization $z = e^{i\theta}$ gives $dz = ie^{i\theta}d\theta = iz\,d\theta$, and we have

$$d\theta \;=\; \frac{dz}{iz}.$$

The trigonometric functions are easily expressible in terms of z on the unit circle. For $\cos\theta$ and $\sin\theta$ on the unit circle we have

$$\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2} = \frac{z+1/z}{2},$$
$$\sin\theta = \frac{e^{i\theta}-e^{-i\theta}}{2i} = \frac{z-1/z}{2i}.$$

If we substitute the expression for $\cos\theta$ into the integral (3.1), we obtain

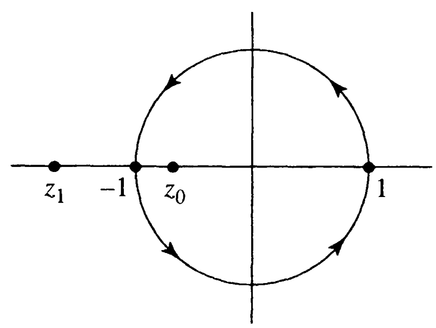
$$\int_0^{2\pi} \frac{d\theta}{a+\cos\theta} = \oint_{|z|=1} \frac{1}{a+\frac{1}{2}(z+1/z)}\frac{dz}{iz} = \frac{2}{i}\oint_{|z|=1}\frac{dz}{z^2+2az+1}.$$

The poles of the integrand are the two zeros of $z^2+2az+1$, which are $-a\pm\sqrt{a^2-1}$. Only one of these roots is inside the unit circle, at $z_0 = -a+\sqrt{a^2-1}$. The residue at z_0 is calculated by Rule 3 above to be

$$\operatorname{Res}\left[\frac{1}{z^2+2az+1}, z_0\right] = \left.\frac{1}{2z+2a}\right|_{z=z_0} = \frac{1}{2\sqrt{a^2-1}}.$$

Thus by the residue theorem

(3.2) $$\int_0^{2\pi}\frac{d\theta}{a+\cos\theta} = \frac{2}{i}\cdot 2\pi i\cdot\frac{1}{2\sqrt{a^2-1}} = \frac{2\pi}{\sqrt{a^2-1}}.$$



unit circle contour

Now consider the integral (3.1) with a replaced by a complex parameter w. We claim that

(3.3) $$\int_0^{2\pi}\frac{d\theta}{w+\cos\theta} = \frac{2\pi}{\sqrt{w^2-1}}, \qquad w\in\mathbb{C}\backslash[-1,1],$$

where the right-hand side of the identity corresponds to the branch of $\sqrt{w^2-1}$ that is positive on the real interval $(1,+\infty)$. Indeed, the functions appearing on each side of the identity (3.3) are analytic for w in the slit plane $\mathbb{C}\backslash[-1,1]$. The identity (3.2) shows that these two analytic functions coincide for $w\in(1,+\infty)$. By the uniqueness principle, the two analytic functions coincide for all $w\in\mathbb{C}\backslash[-1,1]$. This establishes (3.3) also for complex values of the parameter w, except for w in the interval $[-1,1]$, where the integral does not converge.

Exercises for VII.3

1. Show using residue theory that

$$\int_0^{2\pi} \frac{\cos\theta}{2+\cos\theta}\,d\theta \;=\; 2\pi\left(1-\frac{2}{\sqrt{3}}\right).$$

2. Show using residue theory that

$$\int_0^{2\pi} \frac{d\theta}{a+b\sin\theta} \;=\; \frac{2\pi}{\sqrt{a^2-b^2}}\,, \qquad a>b>0.$$

3. Show using residue theory that

$$\int_0^{\pi} \frac{\sin^2\theta}{a+\cos\theta}\,d\theta \;=\; \pi[a-\sqrt{a^2-1}]\,, \qquad a>1.$$

4. Show using residue theory that

$$\int_{-\pi}^{\pi} \frac{d\theta}{1+\sin^2\theta} \;=\; \pi\sqrt{2}.$$

5. Show using residue theory that

$$\int_{-\pi}^{\pi} \frac{1}{1-2r\cos\theta+r^2}\,\frac{d\theta}{2\pi} \;=\; 1, \qquad 0\le r<1.$$

Remark. The integrand is the Poisson kernel (Section X.1).

6. By expanding both sides of the identity

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{d\theta}{w+\cos\theta} \;=\; \frac{1}{\sqrt{w^2-1}}$$

in a power series at ∞, show that

$$\frac{1}{2\pi}\int_0^{2\pi} \cos^{2k}\theta\,d\theta \;=\; \frac{(2k)!}{2^{2k}(k!)^2}\,, \qquad k\ge 0.$$

7. Show using residue theory that

$$\int_0^{2\pi} \frac{d\theta}{(w+\cos\theta)^2} \;=\; \frac{2\pi w}{(w^2-1)^{3/2}}\,, \qquad w\in\mathbb{C}\backslash[-1,1].$$

Specify carefully the branch of the power function. Check your answer by differentiating the integral of $1/(w+\cos\theta)$ with respect to the parameter w.

4. Integrands with Branch Points

Integrals featuring x^a and $\log x$ can sometimes be evaluated using contour integration. It is important to specify carefully the branch of the function z^a or $\log z$ used in the complex integral. We illustrate by deriving the identity

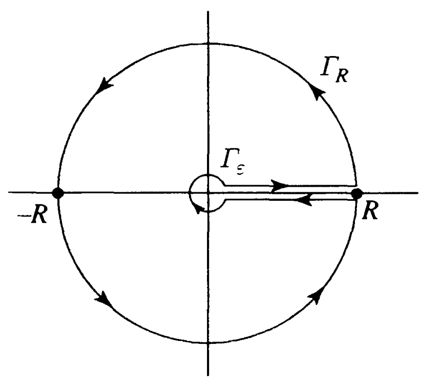
$$(4.1) \qquad \int_0^\infty \frac{x^a}{(1+x)^2}\,dx = \frac{\pi a}{\sin(\pi a)}, \qquad -1 < a < 1.$$

This integral is easily seen to be 1 if $a = 0$, and we interpret the right-hand side also to be 1 at $a = 0$. We suppose then that $a \neq 0$.

We consider the branch of the function $z^a/(1+z)^2$ defined on the slit plane $\mathbb{C}\backslash[0,+\infty)$ by

$$(4.2) \qquad f(z) = \frac{r^a e^{ia\theta}}{(1+z)^2}, \qquad z = re^{i\theta},\ 0 < \theta < 2\pi.$$

We regard the slit $[0,+\infty)$ as having a top edge and a bottom edge, and we extend the function by continuity to each edge of the slit, so it is defined by (4.2) with $\theta = 0$ on the top edge, and by (4.2) with $\theta = 2\pi$ on the bottom edge. The values on the bottom edge are obtained from those on the top edge by multiplying by the phase factor $e^{2\pi i a}$.



keyhole contour

For $\varepsilon > 0$ small and $R > 0$ large, we consider the keyhole domain D consisting of z in the slit plane $\mathbb{C}\backslash[0,\infty)$ satisfying $\varepsilon < |z| < R$. The function $f(z)$ has one pole in D, a double pole at $z = -1$. For the residue, Rule 2 gives

$$\operatorname{Res}\left[\frac{z^a}{(1+z)^2}, -1\right] = \frac{d}{dz}z^a\bigg|_{z=-1} = a\frac{z^a}{z}\bigg|_{z=-1} = -ae^{\pi i a}.$$

The residue theorem yields

$$(4.3) \qquad \int_{\partial D} f(z)\,dz = -2\pi i a e^{\pi i a}.$$

The integral around ∂D breaks into the sum of four integrals, from ε to R along the top edge of the slit, around the circular contour Γ_R of radius R

in the counterclockwise direction, from R back to ε along the bottom edge of the slit, and around the circular contour γ_ε of radius ε in the clockwise direction,

$$\int_\varepsilon^R \frac{x^a}{(1+x)^2}\,dx \;+\; \int_{\Gamma_R} f(z)\,dz \;+\; \int_R^\varepsilon \frac{e^{2\pi i a}x^a}{(1+x)^2}\,dx \;+\; \int_{\gamma_\varepsilon} f(z)\,dz\,.$$

For the integrals over Γ_R and γ_ε, the ML-estimate gives

$$\left|\int_{\Gamma_R} \frac{z^a}{(1+z)^2}\,dz\right| \;\le\; \frac{R^a}{(R-1)^2}\cdot 2\pi R \;\sim\; R^{a-1},$$

$$\left|\int_{\gamma_\varepsilon} \frac{z^a}{(1+z)^2}\,dz\right| \;\le\; \frac{\varepsilon^a}{(1-\varepsilon)^2}\cdot 2\pi\varepsilon \;\sim\; \varepsilon^{a+1}.$$

Since $-1 < a < 1$, both these integrals tend to 0 as $R \to \infty$ and $\varepsilon \to 0$. If we reverse the direction of the integral from R to ε and use (4.3), we obtain in the limit

$$-2\pi i a e^{\pi i a} \;=\; (1 - e^{2\pi i a})\int_0^\infty \frac{x^a}{(1+x)^2}\,dx.$$

This yields the required identity,

$$\int_0^\infty \frac{x^a}{(1+x)^2}\,dx \;=\; \frac{-2\pi i a e^{\pi i a}}{1 - e^{2\pi i a}} \;=\; \frac{2\pi i a}{e^{\pi i a} - e^{-\pi i a}} \;=\; \frac{\pi a}{\sin(\pi a)}\,.$$

This identity can be extended to complex values of the parameter a. The function

$$g(w) \;=\; \int_0^\infty \frac{x^w}{(1+x)^2}\,dx$$

is analytic on the strip $\{-1 < \mathrm{Re}(w) < 1\}$, as is the function $\pi w/\sin(\pi w)$. We have shown that these two functions coincide on the interval $(-1,1)$ where the strip meets the real axis. The uniqueness principle then shows that the functions coincide in the entire strip; that is, the identity (4.1) holds for all complex values of the parameter a satisfying $-1 < \mathrm{Re}(a) < 1$.

Exercises for VII.4

1. By integrating around the keyhole contour, show that

$$\int_0^\infty \frac{x^{-a}}{1+x}\,dx \;=\; \frac{\pi}{\sin(\pi a)}\,, \qquad 0 < a < 1.$$

2. By integrating around the boundary of a pie-slice domain of aperture $2\pi/b$, show that

$$\int_0^\infty \frac{dx}{1+x^b} \;=\; \frac{\pi}{b\sin(\pi/b)}\,, \qquad b > 1.$$

Remark. Check the result by changing variable and comparing with Exercise 1.

3. By integrating around the keyhole contour, show that

$$\int_0^\infty \frac{\log x}{x^a(x+1)}\,dx \;=\; \frac{\pi^2\cos(\pi a)}{\sin^2(\pi a)}, \qquad 0<a<1.$$

Remark. Check the result by differentiating the identity in Exercise 1.

4. For fixed $m \geq 2$, show by integrating around the keyhole contour that

$$\int_0^\infty \frac{x^{-a}}{(1+x)^m}\,dx \;=\; \frac{\pi a(a+1)\cdots(a+m-2)}{(m-1)!\,\sin(\pi a)}, \qquad 1-m<a<1$$

Remark. The result can be obtained also by integrating the formula in Exercise 1 by parts.

5. By integrating a branch of $\dfrac{(\log z)^2}{(z+a)(z+b)}$ around the keyhole contour, show that

$$\int_0^\infty \frac{\log x}{(x+a)(x+b)}\,dx \;=\; \frac{(\log a)^2-(\log b)^2}{2(a-b)}, \qquad a,b>0,\ a\neq b.$$

6. Using residue theory, show that

$$\int_0^\infty \frac{x^a\log x}{(1+x)^2}\,dx \;=\; \frac{\pi\sin(\pi a)-a\pi^2\cos(\pi a)}{\sin^2(\pi a)}, \qquad -1<a<1.$$

7. Show that

$$\int_0^\infty \frac{x^{a-1}}{1+x^b}\,dx \;=\; \frac{\pi}{b\sin(\pi a/b)}, \qquad 0<a<b.$$

Determine for which complex values of the parameter a the integral exists (in the sense that the integral of the absolute value is finite), and evaluate it. Where does the integral depend analytically on the parameter a?

8. By integrating a branch of $(\log z)/(z^3+1)$ around the boundary of an indented sector of aperture $2\pi/3$, show that

$$\int_0^\infty \frac{\log x}{x^3+1}\,dx \;=\; \frac{-2\pi^2}{27}, \qquad \int_0^\infty \frac{1}{x^3+1}\,dx \;=\; \frac{2\pi}{3\sqrt{3}}.$$

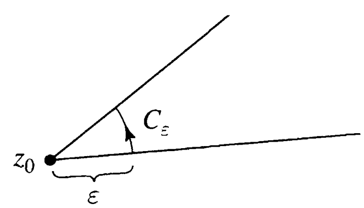
Remark. Compare the results with those of Exercise 3 (after changing variable) and Exercise 2.

9. By integrating around an appropriate contour and using the results of Exercise 8, show that
$$\int_0^\infty \frac{(\log x)^2}{x^3+1}\,dx \;=\; \frac{10\pi^3}{81\sqrt{3}}.$$

10. By integrating a branch of $(\log z)/(z^3-1)$ around the boundary of an indented half-disk and using the result of Exercise 8, show that
$$\int_0^\infty \frac{\log x}{x^3-1}\,dx \;=\; \frac{4\pi^2}{27}.$$

5. Fractional Residues

Suppose z_0 is an isolated singularity of $f(z)$. For $\varepsilon > 0$ small, we consider the integral
$$\int_{C_\varepsilon} f(z)\,dz\,,$$
where C_ε is the arc of the circle $\{|z-z_0| = \varepsilon\}$ subtended by a sector of aperture α. If $\alpha = 2\pi$, then C_ε is the full circle, and the integral is equal to $2\pi i\ \mathrm{Res}[f(z), z_0]$. In general, there is no exact formula for the integral. However, if the isolated singularity of $f(z)$ is a simple pole, then we can calculate the limit of the integrals as $\varepsilon \to 0$.



Theorem (Fractional Residue Theorem). *If z_0 is a simple pole of $f(z)$, and C_ε is an arc of the circle $\{|z-z_0| = \varepsilon\}$ of angle α, then*
$$\lim_{\varepsilon\to 0} \int_{C_\varepsilon} f(z)\,dz \;=\; \alpha i\ \mathrm{Res}[\,f(z)\,,\, z_0\,]\,. \tag{5.1}$$

To see this, express $f(z) = A/(z-z_0)+g(z)$, where A is the residue of $f(z)$ at z_0 and $g(z)$ is analytic at z_0. Parameterizing the circle by $z = z_0 + \varepsilon e^{i\theta}$, and supposing $\theta_0 < \theta < \theta_0 + \alpha$ on the arc, we calculate
$$\int_{C_\varepsilon} \frac{A\,dz}{z-z_0} \;=\; iA \int_{\theta_0}^{\theta_0+\alpha} d\theta \;=\; \alpha i A.$$
Since $g(z)$ is bounded near z_0 and the length of C_ε is at most $2\pi\varepsilon$, the ML-estimate shows that $\int_{C_\varepsilon} g(z)dz$ tends to 0 as $\varepsilon \to 0$. Consequently, $\int_{C_\varepsilon} f(z)dz$ tends to $\alpha i A$ as $\varepsilon \to 0$, which is (5.1).

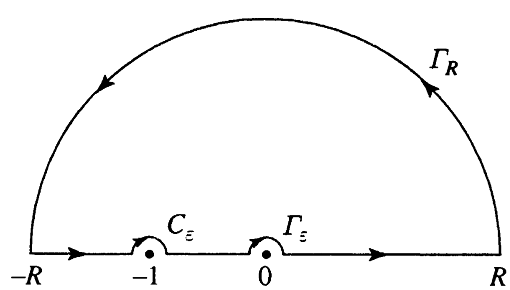
Note that the angle α is taken to be negative if the arc of the circle is traversed in the negative direction, that is, in the clockwise direction.

To illustrate how the fractional residue theorem is used, we show that

$$\int_0^\infty \frac{\log x}{x^2-1}\,dx \;=\; \frac{\pi^2}{4}. \tag{5.2}$$

We choose the branch of $\log z$ that is real on the positive real axis (the principal branch of $\log z$), and we integrate $f(z) = (\log z)/(z^2-1)$ around the boundary ∂D of a domain D that is a half-disk of radius R in the upper half-plane, with indentations of radii ε and δ at the singularities of $f(z)$ at -1 and 0, respectively. Since the zeros of $\log z$ and z^2-1 at $z=1$ are both simple, they cancel each other out, so that $f(z)$ is analytic at $z=1$ and no indentation is required there. Since $f(z)$ is analytic on D, Cauchy's theorem yields

$$\int_{\partial D} f(z)\,dz \;=\; 0. \tag{5.3}$$



indented contour

Now the integral around ∂D breaks into the sum of six integrals, from δ to R along the positive real axis, around the semicircular contour Γ_R of radius R in the counterclockwise direction, from $-R$ to $-1-\varepsilon$ along the negative real axis, around a semicircular contour C_ε of radius ε in the clockwise direction around -1, from $-1+\varepsilon$ to $-\delta$ along the negative real axis, and finally around a semicircular contour γ_δ of radius δ in the clockwise direction around 0. For the integrals over Γ_R and γ_δ, the ML-estimate gives

$$\left|\int_{\Gamma_R} \frac{\log z}{z^2-1}\,dz\right| \;\le\; \frac{\sqrt{\log^2 R+\pi^2}}{R^2-1}\cdot \pi R \;\sim\; \frac{\log R}{R},$$

$$\left|\int_{\gamma_\delta} \frac{\log z}{z^2-1}\,dz\right| \;\le\; \frac{\sqrt{\log^2 \delta+\pi^2}}{1-\delta^2}\cdot \pi\delta \;\sim\; \delta|\log\delta|.$$

If we let $R \to \infty$ and $\delta \to 0$, these two contributions disappear, and (5.3) becomes

$$\int_0^\infty \frac{\log x}{x^2-1}\,dx + \int_{-\infty}^{-1-\varepsilon} \frac{\log|x|+\pi i}{x^2-1}\,dx + \int_{-1+\varepsilon}^{0} \frac{\log|x|+\pi i}{x^2-1}\,dx + \int_{C_\varepsilon} \frac{\log z}{z^2-1}\,dz = 0. \tag{5.4}$$

The integral around C_ε is handled by the fractional residue formula. The angle α is $-\pi$, since C_ε is traversed halfway around the circle in the negative direction. The function $f(z)$ has a simple pole at $z=-1$, with residue computed by Rule 3 to be

$$\operatorname{Res}\left[\frac{\log z}{z^2-1}, -1\right] = \left.\frac{\log z}{2z}\right|_{z=-1} = -\frac{1}{2}\log(-1) = -\frac{i\pi}{2}.$$

Thus the fractional residue formula (5.1) becomes

$$\lim_{\varepsilon\to 0}\int_{C_\varepsilon} \frac{\log z}{z^2-1}\,dz = i(-\pi)(-\frac{i\pi}{2}) = -\frac{\pi^2}{2}.$$

If we take the real part of (5.4) and let $\varepsilon \to 0$, we then obtain

$$\int_0^\infty \frac{\log x}{x^2-1}\,dx + \int_{-\infty}^{0} \frac{\log|x|}{x^2-1}\,dx - \frac{\pi^2}{2} = 0.$$

After a change in variable $x \mapsto -x$ in the second integral, this becomes (5.2).

Exercises for VII.5

1. Use the keyhole contour indented on the lower edge of the axis at $x=1$ to show that

$$\int_0^\infty \frac{\log x}{x^a(x-1)}\,dx = \frac{2\pi^2}{1-\cos(2\pi a)}, \qquad 0<a<1.$$

2. Show using residue theory that

$$\int_{-\infty}^\infty \frac{\sin(ax)}{x(x^2+1)}\,dx = \pi(1-e^{-a}), \qquad a>0.$$

Hint. Replace $\sin(az)$ by e^{iaz}, and integrate around the boundary of a half-disk indented at $z=0$.

3. Show using residue theory that

$$\int_{-\infty}^\infty \frac{\sin(ax)}{x(\pi^2-a^2x^2)}\,dx = \frac{2}{\pi}, \qquad a>0.$$

4. Show using residue theory that

$$\int_0^\infty \frac{1-\cos x}{x^2}\,dx = \frac{\pi}{2}.$$

5. By integrating $(e^{\pm 2iz} - 1)/z^2$ over appropriate indented contours and using Cauchy's theorem, show that

$$\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2}\, dx \;=\; \pi.$$

6. By integrating a branch of $(\log z)/(z^3 - 1)$ around the boundary of an indented sector of aperture $2\pi/3$, show that

$$\int_{0}^{\infty} \frac{\log x}{x^3 - 1}\, dx \;=\; \frac{4\pi^2}{27}.$$

Remark. See also Exercise 4.10.

6. Principal Values

An integral $\int_a^b f(x)dx$ is **absolutely convergent** if the (proper or improper) integral $\int_a^b |f(x)|dx$ is finite. The integral is **absolutely divergent** if $\int_a^b |f(x)|dx = +\infty$. There is essentially only one way to assign a value to an absolutely convergent integral, while there may not be an obvious way to assign a value to an absolutely divergent integral. This is analogous to the dichotomy between absolutely and conditionally convergent series. Every rearrangement of an absolutely convergent series converges to the same value, while the rearrangements of a conditionally convergent series can converge to just about anything.

Example. The integral

$$\int_{-1}^{+1} \frac{1}{x}\, dx$$

is absolutely divergent, since $\int_{-1}^{+1}(1/|x|)dx = +\infty$. One natural way to assign a value to the integral is to take a limit as $\varepsilon \to 0$ of integrals over the two intervals $[-1, -\varepsilon]$ and $[\varepsilon, +1]$, which are obtained by excising a symmetric interval centered at the singularity 0. The negative contributions for $x < 0$ cancel out the positive contributions for $x > 0$, and we obtain what is called the "principal value" of the integral,

$$\mathrm{PV}\int_{-1}^{+1} \frac{1}{x}\, dx \;=\; \lim_{\varepsilon\to 0}\left(\int_{-1}^{-\varepsilon} + \int_{\varepsilon}^{1}\right)\frac{1}{x}\, dx \;=\; 0.$$

Observe, though, that if we excise an asymmetric interval, say from $-\varepsilon$ to $c\varepsilon$, and take a limit, instead of 0 we obtain

$$\lim_{\varepsilon\to 0}\left(\int_{-1}^{-\varepsilon} + \int_{c\varepsilon}^{1}\right)\frac{1}{x}\, dx \;=\; -\log c,$$

which can be any value at all.